

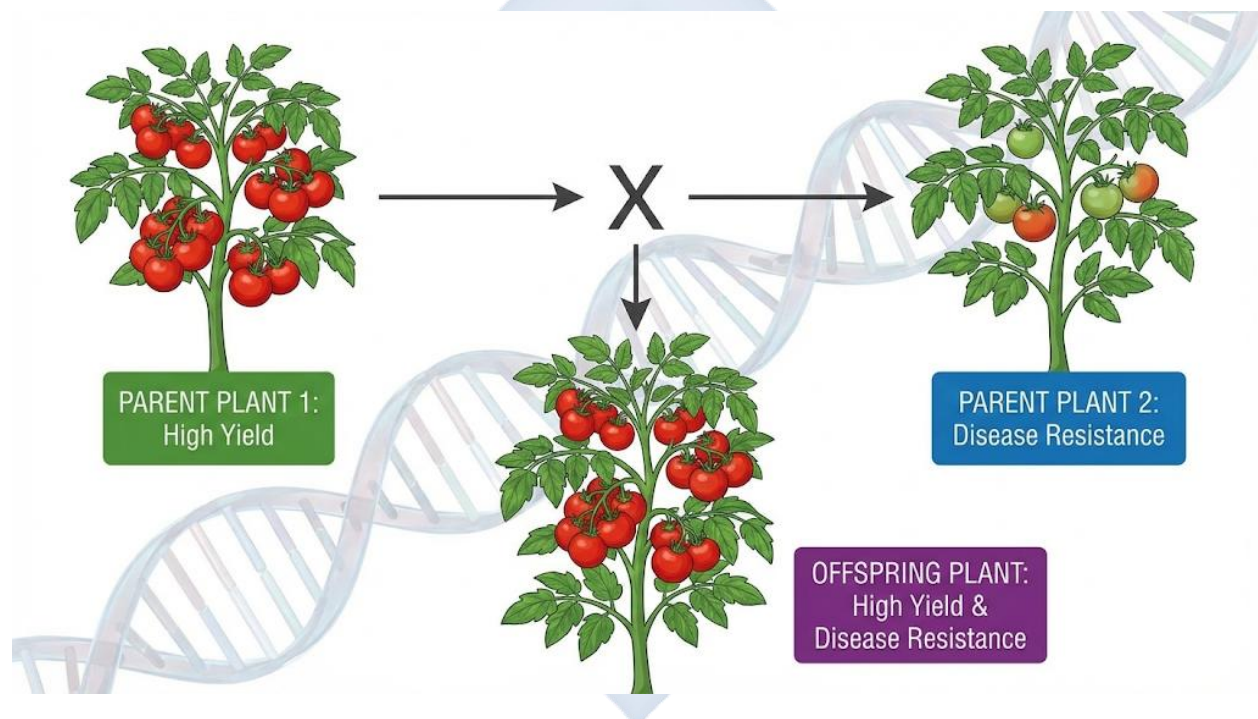
## SS2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

### FIRST TERM

#### WEEK 1: HORTICULTURAL CROP IMPROVEMENT

##### 1.1 INTRODUCTION

Horticultural crop improvement, also known as plant breeding, is the art and science of changing the genetic traits of plants to develop new varieties with more desirable characteristics. It is a deliberate, human-driven evolution of crops to better suit human needs.



##### 1.2 AIMS OF CROP IMPROVEMENT

The primary goal is to make plants more useful and productive. Specific aims include:

1. **Increased Yield:** Developing crops that produce more food, fibre, or ornamental value per unit area (e.g., tomatoes with more fruits per plant, maize with larger cobs).
2. **Improved Quality:**

- **Nutritional Quality:** Enhancing vitamin, protein, or mineral content (e.g., "Golden Rice" enriched with Vitamin A).
  - **Eating Quality:** Better taste, flavour, tenderness, or texture.
  - **Aesthetic Quality:** Brighter colours, larger flowers, or more attractive shapes for ornamental plants.
  - **Keeping Quality:** Longer shelf-life for fruits and vegetables, reducing post-harvest losses.
3. **Pest and Disease Resistance:** Breeding plants that can naturally defend themselves against specific insects, fungi, bacteria, or viruses (e.g., cassava resistant to Cassava Mosaic Disease).
  4. **Abiotic Stress Tolerance:** Developing "hardy" crops that can withstand difficult environmental conditions like:
    - Drought tolerance (e.g., maize that survives with less water).
    - Salinity tolerance (for coastal or irrigated areas).
    - Heat or cold tolerance.
  5. **Desirable Plant Architecture:** Modifying the plant's shape or structure (e.g., creating dwarf rice varieties that don't fall over, or climbing bean varieties for vertical gardens).
  6. **Ease of Harvesting:** Breeding crops that mature uniformly for mechanical harvesting or have fewer thorns.

### 1.3 METHODS OF CROP IMPROVEMENT

Plant breeders use several established methods to achieve these aims.

#### 1. SELECTION

This is the oldest and most basic method. It involves identifying plants with desirable traits and using them to create the next generation.

- **Mass Selection:**
  - **Process:** A large number of plants (a "mass") are evaluated in the field. The "best" looking plants (phenotype) are selected and harvested. Their seeds are bulked together and planted to form the next generation.

- **Use:** Common for cross-pollinated crops. It's simple and cheap but slow and not very precise.
- **Pure-line Selection:**
  - **Process:** Used for self-pollinating crops. A breeder selects a *single* superior plant. The seeds from this *one* plant are collected and grown out. Since the plant self-pollinates, all its offspring are genetically very similar (a "pure-line").
  - **Use:** Good for creating highly uniform varieties (e.g., a new tomato or bean variety).
- **Clonal Selection:**
  - **Process:** Used for crops that are propagated vegetatively (asexually) like cassava, yams, bananas, or potatoes. A breeder identifies a single superior plant (e.g., a cassava plant with huge tubers). Cuttings ("clones") are taken from this one plant and multiplied.
  - **Use:** All offspring are genetically identical to the superior parent.

## 2. HYBRIDIZATION (CROSS-BREEDING)

This is the most powerful traditional breeding method. It involves crossing two parent plants with different desirable traits, to produce offspring that combines those traits.

- **Process:**
  1. **Parent Selection:** Choose two parents (Parent A and Parent B).
    - *Example:* Parent A (Tomato) = High-yielding but disease-susceptible.  
Parent B (Tomato) = Low-yielding but disease-resistant.
    - *Goal:* Create a new variety that is **High-yielding AND Disease-resistant**.
  2. **Emasculation:** To prevent self-pollination, the breeder must carefully remove the male parts (stamens/anthers) from the flower of Parent A (the designated "female" parent) before it produces pollen.
  3. **Pollination:** The breeder collects pollen from the flower of Parent B (the "male" parent) and manually transfers it to the stigma (female part) of the emasculated flower of Parent A.

4. **Bagging:** The pollinated flower is covered with a small bag (paper or cloth) to prevent stray, unwanted pollen from landing on it.
  5. **Harvesting F1:** The flower develops into a fruit (a tomato). The seeds inside are the "First Filial" (F1) generation. These seeds contain genes from both parents.
  6. **Growing F1 & F2:** The F1 seeds are planted. The F1 plants are often self-pollinated to produce the "Second Filial" (F2) generation.
  7. **Selection (The long part):** The F2 generation is where all the gene combinations appear. The breeder must plant thousands of F2 seeds and *select* the very few individual plants that show the desired combination (high yield + disease resistance).
  8. **Stabilization:** These selected plants are grown and re-selected over many (6-10) generations until their desirable traits are "fixed" (i.e., they breed true).
- **Types of Hybridization:**
    - **Intraspecific:** Crossing two different varieties *within* the same species (e.g., *Zea mays* 'A' x *Zea mays* 'B'). This is the most common type.
    - **Interspecific:** Crossing two different (but related) *species* (e.g., to transfer a wild, disease-resistant gene into a cultivated species).

## EVALUATION (WEEK 1)

1. Define horticultural crop improvement in your own words.
2. List four distinct aims of a plant breeding programme.
3. Explain the difference between "Pure-line Selection" and "Clonal Selection," stating which type of crop each is used for.
4. What is "Emasculation" in the context of hybridization, and why is it necessary?
5. A farmer has a maize variety that tastes sweet (Trait A) and another that is drought-tolerant (Trait B). Which *method* of crop improvement would be best to combine these two traits?

## ASSIGNMENT (WEEK 1)

1. Visit a local market or farm.

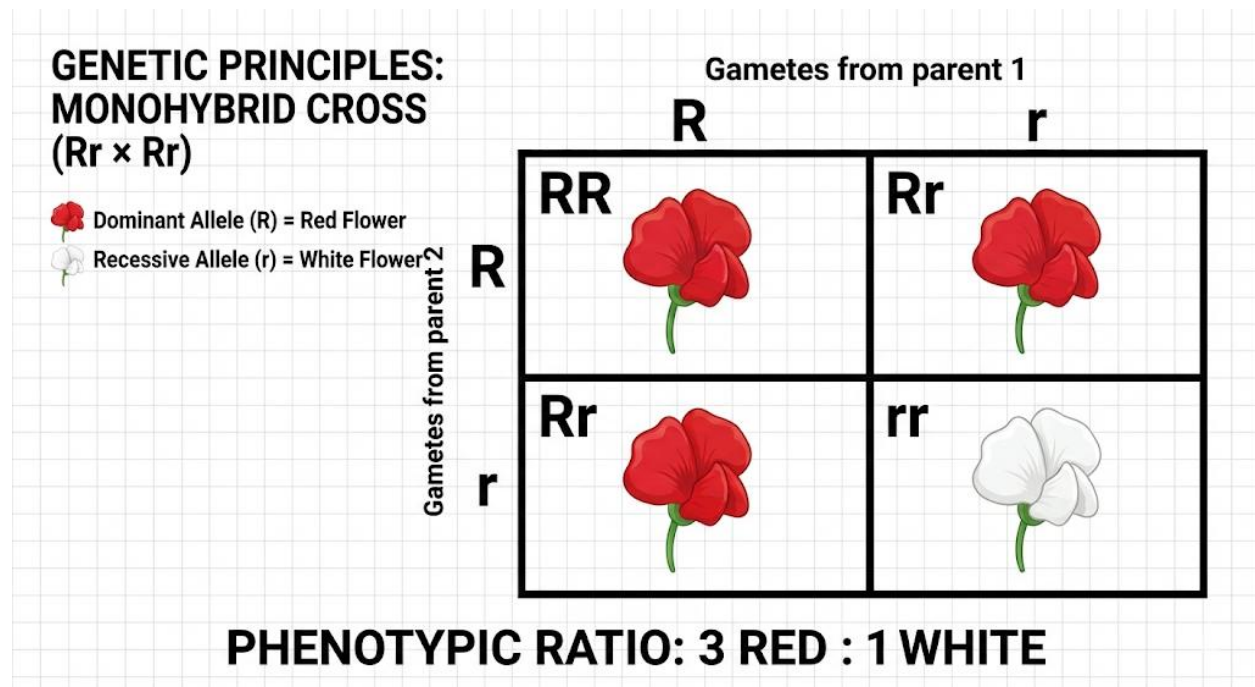
2. Identify two different varieties of the *same* crop (e.g., two types of mango, two types of pepper, or two types of cassava - yellow vs. white).
3. In your notebook, list the observable differences (phenotype) between them (e.g., size, colour, shape, texture).
4. Write a short paragraph speculating on what "aims" a breeder might have had when developing one of the varieties.



## WEEK 2: GENETIC PRINCIPLES IN CROP PRODUCTION

### 2.1 INTRODUCTION

Genetics is the science of heredity. It explains *how* traits are passed from parents to offspring. This is the "science" behind the "art" of plant breeding. The foundation of this science was laid by Gregor Mendel.



### 2.2 KEY GENETIC TERMINOLOGY

- **Gene:** The basic unit of heredity; a segment of DNA that codes for a specific trait (e.g., the gene for flower colour).
- **Allele:** Different forms of the same gene (e.g., for the flower colour gene, the alleles might be 'R' for red and 'r' for white).
- **Phenotype:** The *physical appearance* or observable trait of an organism (e.g., "The flower is red").
- **Genotype:** The *genetic makeup* of an organism; the combination of alleles it has (e.g., 'RR', 'Rr', or 'rr').



- **Homozygous:** Having two identical alleles for a trait (e.g., **RR** or **rr**). This is "true-breeding."
- **Heterozygous:** Having two different alleles for a trait (e.g., **Rr**).
- **Dominant Allele:** An allele that *masks* the effect of a recessive allele when present (e.g., if 'R' is dominant, both **RR** and **Rr** genotypes will have red flowers). Represented by a capital letter.
- **Recessive Allele:** An allele that is *masked* by a dominant allele. It is only expressed in the phenotype when it is homozygous (e.g., **rr** for white flowers). Represented by a small letter.

## 2.3 MENDEL'S LAWS OF HEREDITY

### 1. MENDEL'S FIRST LAW: THE LAW OF SEGREGATION

- **Concept:** For any trait, an organism inherits two alleles, one from each parent. These two alleles *segregate* (separate) from each other during the formation of gametes (pollen or ovules), so that each gamete contains only *one* allele for that trait.
- **Example (Monohybrid Cross):**
  - A homozygous dominant plant (**RR** - Red) is crossed with a homozygous recessive plant (**rr** - white).
  - **P (Parent) Generation:** **RR** x **rr**
  - **Gametes:** The **RR** parent only makes **R** gametes. The **rr** parent only makes **r** gametes.
  - **F1 Generation:** All offspring receive one 'R' and one 'r', so all are **Rr** (genotype) and all have Red flowers (phenotype).
  - Now, cross two F1 plants: **Rr** x **Rr**
  - **Gametes:** Each F1 parent produces gametes that are 50% **R** and 50% **r**.
  - **F2 Generation (Using a Punnett Square):**

|          |           |
|----------|-----------|
| <b>R</b> | <b>r</b>  |
| <b>R</b> | <b>Rr</b> |
| <b>r</b> | <b>rr</b> |
  - **Phenotypic Ratio:** 3 Red : 1 White (75% Red, 25% White)
  - **Genotypic Ratio:** 1 **RR** : 2 **Rr** : 1 **rr** (25% Homozygous Dominant, 50% Heterozygous, 25% Homozygous Recessive)

- **Significance:** This law explains why traits can "disappear" in the F1 generation (white) and "reappear" in the F2 generation.

## 2. MENDEL'S SECOND LAW: THE LAW OF INDEPENDENT ASSORTMENT

- **Concept:** Alleles for *different* traits are inherited *independently* of one another. The inheritance of one trait (like flower colour) does not affect the inheritance of another trait (like plant height).
- **Example (Dihybrid Cross):**
  - A parent that is heterozygous for two traits: Round, Yellow seeds (**RrYy**)
  - The gene for shape (R/r) and the gene for colour (Y/y) are on different chromosomes.
  - When this plant makes gametes, the alleles "assort independently." It can make four types of gametes: **RY, Ry, rY, ry**.
  - When two such plants are crossed (RrYy x RrYy), it results in the classic **9:3:3:1 phenotypic ratio** (9 Round-Yellow, 3 Round-green, 3 wrinkled-Yellow, 1 wrinkled-green).
- **Significance:** This is the *scientific reason* why hybridization (Week 1) works! It shows that we *can* create new combinations of traits. We can take "Round seeds" from one parent and "Yellow colour" from another and combine them.

### 2.4 APPLICATIONS IN CROP PRODUCTION

1. **Systematic Breeding:** Mendel's laws turned plant breeding from guesswork into a predictive science. Breeders can now use Punnett squares to predict the outcomes of their crosses.
2. **Hybrid Vigour (Heterosis):** This is a key application. It is the phenomenon where the F1 hybrid (e.g., **Rr**) is superior to *both* of its homozygous parents (e.g., **RR** and **rr**). It might be bigger, more disease-resistant, or higher-yielding. The entire hybrid seed industry (e.g., hybrid maize) is built on this principle. Farmers buy F1 seeds, get the hybrid vigour, but *cannot* save the seeds, as the F2 generation will segregate and lose the superior traits.



3. **Understanding Recessive Traits:** Breeders can now understand why a undesirable trait (like susceptibility to a disease) might be "hidden" as a recessive allele in a heterozygous plant.

### **EVALUATION (WEEK 2)**

1. Differentiate between "Genotype" and "Phenotype" using a clear example.
2. What does it mean for an organism to be "heterozygous" for a trait?
3. State Mendel's Law of Segregation.
4. In tomato plants, red fruit (R) is dominant over yellow fruit (r). You cross a heterozygous (Rr) plant with a yellow-fruited (rr) plant. What is the expected phenotypic ratio of the offspring? (Use a Punnett square to show your work).
5. What is "Hybrid Vigour" (Heterosis), and why is it important in commercial agriculture?

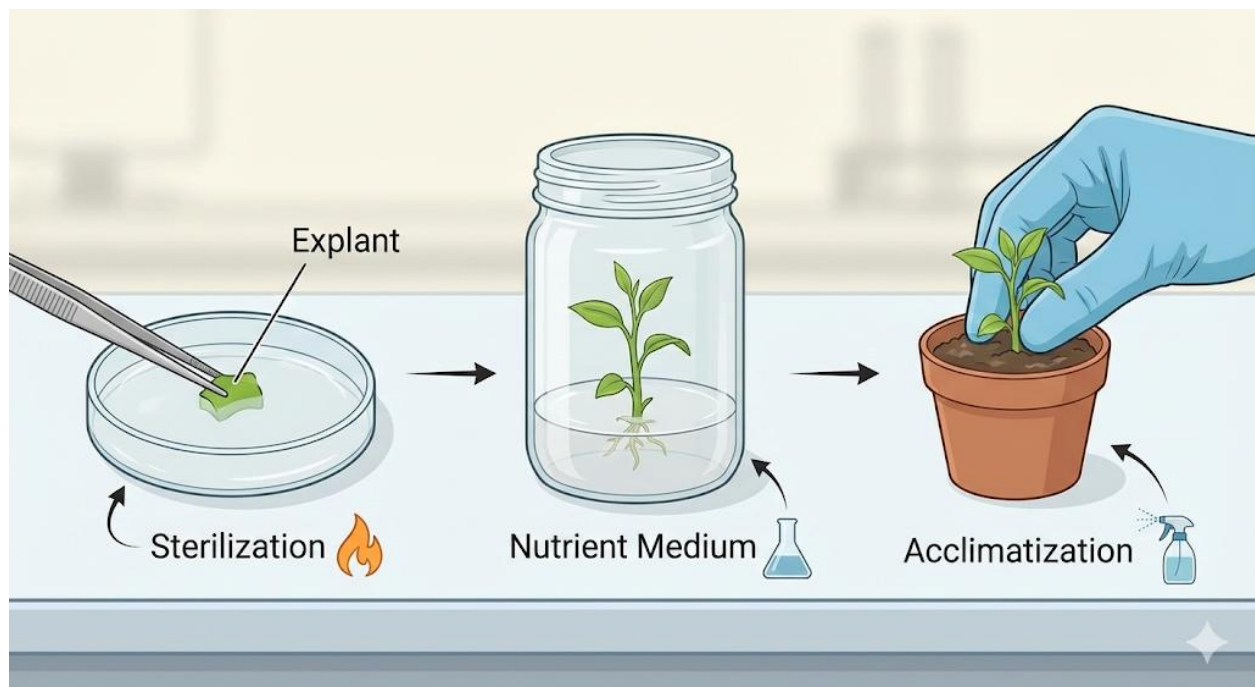
### **ASSIGNMENT (WEEK 2)**

1. Define "Dihybrid Cross."
2. Draw the 16-box Punnett square for a Dihybrid Cross between two parents that are heterozygous for both Round (R) and Yellow (Y) seeds (RrYy x RrYy).
3. Count and list the four different phenotypes and their 9:3:3:1 ratio.

## WEEK 3: BIOTECHNOLOGY IN HORTICULTURE

### 3.1 INTRODUCTION

Biotechnology involves using living organisms or their products to make or modify products for a specific use. In horticulture, it allows for *rapid* and *precise* plant improvement, far beyond the speed of traditional breeding.



### 3.2 TISSUE CULTURE (MICROPROPAGATION)

#### A. DEFINITION

Tissue culture is an *in-vitro* (in glass) technique of growing new plants from tiny pieces of plant tissue (like a leaf, stem, or root tip) on a sterile, artificial nutrient medium (like a jelly). The small piece of tissue used is called an **explant**. The key principle is **totipotency** – the ability of a single plant cell to grow into a complete, new plant.

#### B. THE PROCESS

1. **Explant Selection:** A small, healthy piece of the desired plant (the "mother plant") is selected (e.g., the meristem tip, a leaf section, a bud).
2. **Sterilization:** This is the most critical step. The explant is surface-sterilized (using bleach or alcohol) to kill all fungi and bacteria. The nutrient medium and all tools are also sterilized (usually in an autoclave). The work is done in a **laminar flow hood** which provides sterile air.
3. **Inoculation:** The sterile explant is placed onto the sterile nutrient medium (a gel containing water, sugar, macro/micro-nutrients, and vitamins) in a test tube or jar.
4. **Incubation:** The jar is placed in a growth room with controlled light and temperature.
5. **Callus & Shoot Formation:** The explant cells divide and form an undifferentiated mass of cells called a **callus**. By adding specific **plant hormones** (auxins and cytokinins) to the medium, the breeder can encourage the callus to form shoots.
6. **Rooting:** The new, small shoots are transferred to a different medium containing hormones that promote root growth.
7. **Acclimatization (Hardening Off):** The small "plantlets" are removed from the sterile, high-humidity jars. Their roots are washed, and they are planted in sterile soil in a greenhouse. They must be slowly adapted (acclimatized) to the "real world" conditions of lower humidity and natural soil.

### C. APPLICATIONS OF TISSUE CULTURE

1. **Mass Propagation (Cloning):** One explant can produce *thousands or millions* of genetically identical (clonal) plantlets in a short time. This is vital for commercial production of ornamentals (e.g., orchids) and crops like bananas, plantains, and yams.
2. **Production of Disease-Free Plants:** The *meristem tip* (the very tip of a growing shoot) is often free of viruses, even if the mother plant is infected. By tissue-culturing the meristem, "clean" planting material can be produced.
3. **Germplasm Conservation:** Plant varieties can be stored as tiny, slow-growing tissue cultures in a lab ("gene bank"), saving space and protecting against field losses.
4. **Speeding up Breeding:** Allows breeders to grow new hybrids or conduct genetic experiments much faster than in the field.

### 3.3 GENETIC MODIFICATION (GM CROPS)

#### A. DEFINITION

Genetic Modification (GM), or Genetic Engineering (GE), is a biotechnology technique that involves *directly* inserting a specific gene from one organism (e.g., a bacterium, virus, or another plant) into the DNA of another organism (the crop). The resulting plant is a "Genetically Modified Organism" (GMO) or a "transgenic" plant. This is different from hybridization, which can only cross related species.

#### B. THE PROCESS (SIMPLIFIED)

1. **Gene Identification:** A scientist identifies a useful gene (e.g., the Bt gene from the bacterium *Bacillus thuringiensis*, which is toxic to insects).
2. **Gene Isolation:** The gene is isolated ("cut out") using special "restriction" enzymes.
3. **Gene Insertion (Transformation):** The isolated gene is inserted into the plant's cells.

Two common methods are:

- **Gene Gun:** The gene is coated onto tiny gold particles and "shot" into the plant cells.
  - ***Agrobacterium tumefaciens*:** Using a natural soil bacterium that has the ability to transfer its own DNA (on a "Ti plasmid") into a plant. Scientists replace the bacterium's DNA with the desired gene.
4. **Selection & Regeneration:** The cells are grown using tissue culture. The medium contains an "antibiotic marker" so that only the cells that *successfully* took up the new gene will survive. These cells are then regenerated into whole plants.

#### C. APPLICATIONS OF GM CROPS

1. **Insect Resistance:** Bt Cotton and Bt Maize. They produce a protein that kills specific insect pests (like the bollworm), reducing the need for chemical insecticides.
2. **Herbicide Tolerance:** "Roundup Ready" Soybeans or Maize. They are engineered to be resistant to the herbicide glyphosate. This allows a farmer to plant the crop and then spray the entire field with the herbicide, killing the weeds but not the crop.

3. **Nutritional Enhancement:** "Golden Rice." It is engineered to produce beta-carotene, a precursor to Vitamin A, to help combat vitamin deficiency in developing countries.
4. **Disease Resistance:** Engineering plants to be resistant to viruses (e.g., Papaya Ringspot Virus).

#### D. BENEFITS AND CONCERNS

- **Benefits:** Increased yields, reduced pesticide use, improved nutrition, enhanced food security.
- **Concerns (The Controversy):**
  - *Environmental:* Risk of "superweeds" (herbicide resistance spreading) or harm to non-target insects.
  - *Human Health:* Concerns (largely unproven) about long-term safety or allergies.
  - *Economic:* Control of seeds by a few large corporations; farmers cannot save GM seeds.

#### EVALUATION (WEEK 3)

1. What is "totipotency," and why is it the central principle of tissue culture?
2. Describe the process of "Acclimatization" (Hardening Off) and explain why it is necessary.
3. List three major applications (uses) of tissue culture in horticulture.
4. What is the key difference between traditional hybridization and genetic modification (GM)?
5. Explain what Bt Cotton is. Where did the Bt gene come from, and what is the benefit to the farmer?

#### ASSIGNMENT (WEEK 3)

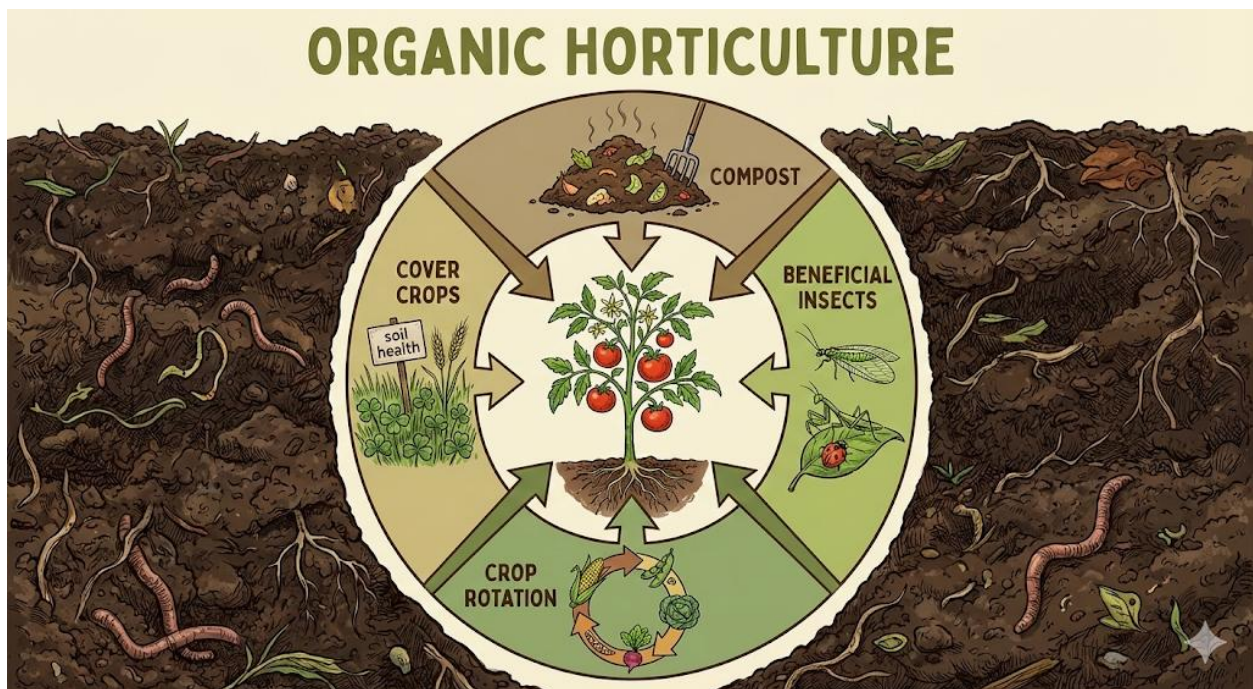
1. Draw a simple flowchart (diagram with arrows) showing the 7 main steps of tissue culture, starting from "Explant Selection" and ending at "Acclimatization."
2. Write a short paragraph explaining the difference between "Herbicide Tolerant" GM crops and "Insect Resistant" GM crops.



## WEEK 4: ORGANIC HORTICULTURE

### 4.1 INTRODUCTION

Organic horticulture is a holistic production system that avoids the use of synthetic (man-made) fertilizers, pesticides, and herbicides. It focuses on building healthy soil and working *with* natural ecological cycles to produce nutritious food.



### 4.2 PRINCIPLES OF ORGANIC HORTICULTURE

The philosophy is guided by four main principles:

1. **Principle of Health:** Aims to produce healthy soil, healthy plants, healthy animals, and healthy people. It avoids inputs that can harm health.
2. **Principle of Ecology:** Works *with* living ecological systems, not against them. It emphasizes recycling, biodiversity, and balance in nature.
3. **Principle of Fairness:** Seeks to ensure fairness in relationships with all involved, from farmers and workers to consumers.



4. **Principle of Care:** Emphasizes caution and responsibility in managing the farm to protect the environment and the well-being of future generations.

#### 4.3 KEY PRACTICES IN ORGANIC HORTICULTURE

Organic farmers replace synthetic inputs with natural, knowledge-based practices.

##### A. SOIL FERTILITY MANAGEMENT

(Goal: "Feed the soil, not the plant.")

1. **Composting:** The controlled decomposition of organic matter (kitchen scraps, leaves, grass clippings, small amounts of manure) by microorganisms. The result, "compost," is a dark, nutrient-rich soil conditioner.
2. **Animal Manures:** Using waste from poultry, goats, or cattle. It *must* be composted or "aged" first, as raw manure can burn plants and contain pathogens.
3. **Green Manures / Cover Cropping:**
  - **Cover Crop:** A crop (like *Mucuna*, *Centrosema*, or *Pueraria*) planted *between* cash crops or during the off-season to cover bare soil.
  - **Green Manure:** A cover crop that is specifically grown to be cut down and tilled *into* the soil to add organic matter. Leguminous cover crops (beans, peas) are special because they "fix" nitrogen from the air into the soil.
4. **Crop Rotation:** Not planting the same crop in the same spot year after year. (e.g., Plant maize (heavy feeder) -> then beans (nitrogen fixer) -> then yams (root crop)). This breaks pest cycles and balances soil nutrient use.

##### B. PEST AND DISEASE CONTROL

(Goal: Not to *eliminate* all pests, but to keep them at a manageable, non-damaging level.)

1. **Cultural Control:**
  - *Crop Rotation:* (As above) Prevents soil-borne diseases from building up.
  - *Intercropping:* Planting two or more different crops together (e.g., maize and cassava). This confuses pests.

- *Sanitation*: Removing and destroying diseased plants and crop debris.
- 2. **Biological Control:**
  - Encouraging "beneficial" insects (like ladybugs, which eat aphids) by planting flowers they like.
  - Using microbial pesticides like *Bacillus thuringiensis* (Bt) – (Note: this is the *natural bacteria*, not the GM crop).
- 3. **Physical/Mechanical Control:**
  - Hand-picking large insects (like caterpillars) off plants.
  - Using insect traps or reflective mulches.
- 4. **Natural Pesticides:**
  - Using plant-based sprays like **Neem Oil** (*Azadirachta indica*), garlic spray, or hot pepper spray. These are biodegradable but must still be used with care.

## C. WEED CONTROL

1. **Mulching**: Covering the soil surface with organic materials (straw, grass clippings, wood chips). This blocks sunlight, preventing weed seeds from germinating, and also conserves soil moisture.
2. **Cover Cropping**: (As above) A dense cover crop will "choke out" and smother weeds.
3. **Hand Weeding / Hoeing**: Mechanical removal of weeds.

## 4.4 BENEFITS AND CHALLENGES

- **Benefits:**
  - *Environmental*: No chemical runoff to pollute water; improves soil health and biodiversity.
  - *Health*: Food is free of synthetic pesticide residues.
  - *Sustainability*: Builds long-term soil fertility.
  - *Economic*: (Potentially) Higher "premium" price for certified organic food; lower cost of inputs (no chemicals).
- **Challenges:**
  - *Labour Intensive*: Weeding and compost-making require more physical work.

- *Lower Yields*: (Sometimes, especially in the transition period).
- *Certification*: The process to become "Certified Organic" can be complex and expensive for small farmers.

#### **EVALUATION (WEEK 4)**

1. What is the core philosophy of organic horticulture regarding soil? ("Feed the...")
2. Explain the difference between a "Cover Crop" and "Green Manure."
3. Why is "Crop Rotation" a key practice in organic farming? (Give two reasons).
4. List three *non-chemical* methods an organic farmer can use to control insect pests.
5. What are two benefits and one challenge of organic horticulture?

#### **ASSIGNMENT (WEEK 4)**

1. Describe how you would make a small compost heap or pile at home or in the school garden.
2. List five "GREEN" materials (Nitrogen-rich, e.g., kitchen scraps, fresh grass) and five "BROWN" materials (Carbon-rich, e.g., dry leaves, cardboard, sawdust) that can be used for compost.

## **WEEK 5: MIDTERM EXAMINATION**

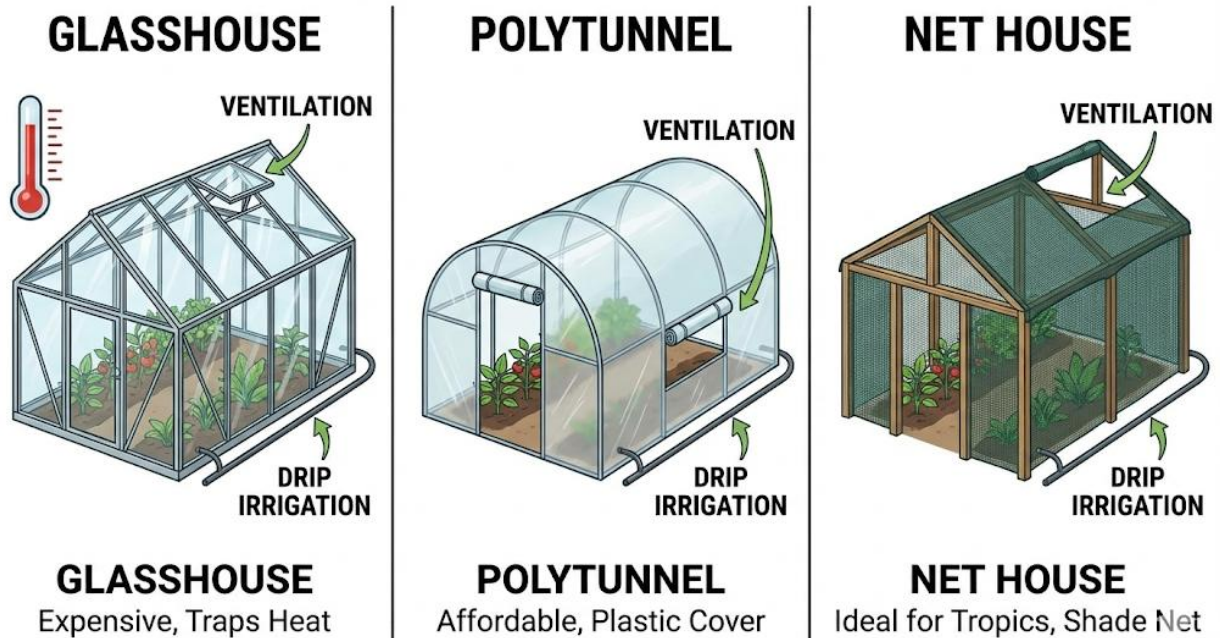
*This week is for assessment of all topics covered in Weeks 1-4.*



## WEEK 6: GREENHOUSE TECHNOLOGY

### 6.1 INTRODUCTION

A greenhouse (or glasshouse) is a structure designed to provide a controlled, protected environment for growing crops. It is covered with a transparent or translucent material (like glass or plastic) that allows sunlight to enter and heats the interior.



### 6.2 STRUCTURES AND TYPES

1. **Glasshouses:** The traditional structure, using glass panels in a metal (aluminum) frame.
  - *Pros:* Very long-lasting (30+ years), best light transmission.
  - *Cons:* Very expensive, heavy, fragile (can break), traps heat (a major problem in Nigeria).
2. **Polytunnels (Hoop Houses):**
  - **Structure:** Arched hoops (metal or PVC) covered with a single or double layer of UV-stabilized polyethylene (polythene) plastic.
  - *Pros:* Much cheaper, easy to build, good for extending seasons.
  - *Cons:* Plastic needs replacing every 3-5 years, can be damaged by wind.

### 3. Net Houses (Shade Houses):

- **Structure:** A simple frame (wood or metal) covered with **shade net** or **insect-proof netting**.
- *Pros:* **Very suitable for Nigerian/tropical climates.** Does *not* trap heat; provides shade (reducing heat stress); and physically *excludes* insect pests (like whiteflies).
- *Cons:* Does not protect from heavy rain.

## 6.3 ENVIRONMENTAL CONTROL (THE "TECHNOLOGY")

A greenhouse allows the grower to become a "weather man" and control key factors:

### 1. Temperature:

- *Cooling (Critical in Nigeria):* **Ventilation** is key. This includes side-vents that roll up, roof vents that open, and large "extraction" fans to pull hot air out. "Evaporative cooling" (using wet pads and fans) is a more advanced method.
- *Heating (For cold climates/nights):* Using electric or gas heaters.

### 2. Light:

- *Reducing:* Using shade cloths (nets) over the roof to reduce intense sunlight.
- *Adding:* Using artificial "grow lights" (e.g., LEDs) to supplement light on cloudy days or extend the "day length."

### 3. Water & Humidity:

- *Irrigation:* **Drip irrigation** (delivering water directly to the plant's roots) is the most common and efficient method.
- *Humidity:* Mistlers or foggers can be used to raise humidity for seedlings or certain crops.

### 4. Pest & Disease Control:

- The structure is a physical barrier.
- **Double-door entry systems** act as an "airlock" to stop insects from flying in.
- Fine insect netting on all vents and openings.

## 6.4 BENEFITS OF GREENHOUSE TECHNOLOGY



1. **Year-Round Production:** Allows farmers to grow crops (especially high-value vegetables like tomatoes, peppers, cucumbers) even in the off-season.
2. **Protection from Elements:** Protects crops from heavy rain (which can spread disease), strong winds, and hailstones.
3. **Pest and Disease Control:** Drastically reduces pest and disease pressure, leading to less pesticide use and higher quality produce.
4. **Increased Yield:** Optimized conditions (water, nutrients) lead to much higher yields per square meter.
5. **Water Efficiency:** Drip irrigation in a greenhouse uses up to 90% less water than traditional field irrigation.

## 6.5 CHALLENGES IN NIGERIA

1. **High Initial Cost:** The setup cost for glasshouses or well-equipped polytunnels is very high.
2. **Overheating:** This is the *number one* challenge. A poorly designed, poorly ventilated greenhouse will become an "oven" in the tropical sun and kill plants.
3. **Constant Power Supply:** Advanced systems (fans, pumps, coolers) require a reliable electricity source, which is a major problem (requiring generators or large solar setups).
4. **Technical Know-How:** Managing a greenhouse environment requires training and skill.
5. **Availability of Materials:** Sourcing high-quality UV-treated plastic, shade nets, and parts can be difficult or expensive.

## EVALUATION (WEEK 6)

1. What is a greenhouse?
2. Compare a "Glasshouse" and a "Polytunnel" in terms of cost and durability.
3. Why is a "Net House" or "Shade House" often more suitable for a tropical climate like Nigeria than a fully enclosed plastic greenhouse?
4. List three environmental factors a grower can control in a greenhouse.
5. What is the *single biggest* challenge of using greenhouse technology in Nigeria, and what is the primary method to combat it?

## **ASSIGNMENT (WEEK 6)**

1. Sketch a simple design for a "Net House" suitable for growing tomatoes.
2. Label the following parts: (a) Frame (wood or metal), (b) Shade net roof, (c) Insect-proof net sides, (d) Drip irrigation lines, (e) Double-door entry.

## **WEEK 7: MIDTERM BREAK**

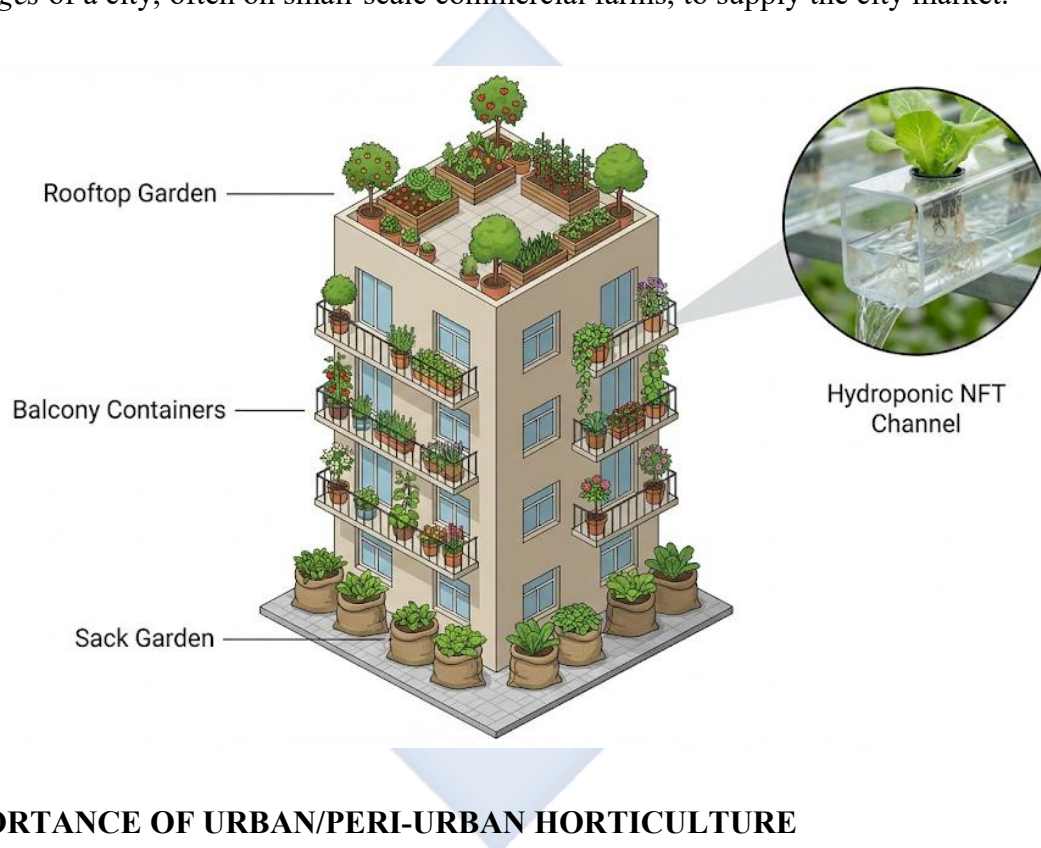
*This is a break from instruction. Students are encouraged to review the first half of the term's work.*



## WEEK 8: URBAN AND PERI-URBAN HORTICULTURE

### 8.1 DEFINITIONS

- **Urban Horticulture:** The practice of growing horticultural crops (fruits, vegetables, flowers) *inside* the city boundaries. This includes backyards, balconies, rooftops, and community gardens.
- **Peri-Urban Horticulture:** The practice of growing crops on the immediate *fringes or edges* of a city, often on small-scale commercial farms, to supply the city market.



### 8.2 IMPORTANCE OF URBAN/PERI-URBAN HORTICULTURE

1. **Food Security & Nutrition:** Provides direct access to fresh, nutritious food (especially vegetables) for city dwellers, improving family diets.
2. **Income Generation:** Creates jobs and business opportunities (selling fresh produce, seedlings, or compost).
3. **Environmental Improvement:**
  - "Greens" the city, improving air quality and reducing the "urban heat island" effect.

- Reduces "Food Miles" (the distance food travels from farm to plate), saving energy and reducing carbon emissions.
  - Waste Recycling: Can utilize composted urban organic waste.
4. **Social & Educational:** Community gardens bring people together. School gardens are powerful teaching tools.

### 8.3 PRACTICES AND SYSTEMS FOR URBAN HORTICULTURE

Since space is the biggest limitation, these methods focus on growing in small, intensive spaces.

1. **Container Gardening:** Growing plants in any available container: plastic pots, buckets, "Grow Bags," old car tires, wooden boxes.
2. **Sack (Bag) Gardening:** Filling large woven sacks (like those for rice or garri) with soil and compost. Plants (like leafy greens) are planted on the top and in holes cut in the sides. Very space-efficient.
3. **Vertical Gardening:** Growing plants *upwards* to save space.
  - Using trellises for climbing plants (beans, cucumbers).
  - Using "stackable" pots or "living wall" systems.
4. **Rooftop Gardens:** Using flat roofs as growing spaces. This requires:
  - Ensuring the roof is strong enough to hold the weight of wet soil.
  - Waterproofing and good drainage to protect the building.
5. **Community Gardens:** A single, larger piece of land (often on "borrowed" land) that is divided into small plots for different families to farm.

### 8.4 HYDROPONICS: SOIL-LESS HORTICULTURE

Hydroponics is an advanced form of urban horticulture that *does not use soil*.

- **Definition:** A method of growing plants in a "nutrient solution" (water mixed with all the essential plant nutrients) or in an inert growing medium (like gravel, perlite, or coconut coir).
- **Common Systems:**

- **Nutrient Film Technique (NFT):** A thin "film" of nutrient water flows continuously in a channel over the bare plant roots. (Common for lettuce).
- **Deep Water Culture (DWC):** Plant roots are suspended directly in a large, aerated (bubbling) tank of nutrient solution.
- **Pros (Advantages):**
  - **No Soil Needed:** Can be done anywhere (a roof, a warehouse, a balcony).
  - **Extremely Fast Growth:** Plants get *exactly* the nutrients they need, 24/7.
  - **Very High Yield:** Produces more food in less space.
  - **Water Efficient:** The water is recycled; uses 70-90% less water than soil farming.
- **Cons (Disadvantages):**
  - **High Initial Cost:** Need for pumps, tanks, plumbing, and nutrient salts.
  - **Technical Skill:** Requires knowledge of chemistry to mix the nutrient solution.
  - **Reliant on Electricity:** The pumps and aerators *must* run 24/7. A power cut can kill the entire crop.

## EVALUATION (WEEK 8)

1. What is the main difference between "Urban" and "Peri-Urban" horticulture?
2. List three (3) important benefits of urban horticulture for a city like Lagos or Abuja.
3. Describe "Sack Gardening."
4. Define Hydroponics.
5. What are two major advantages and two major disadvantages of hydroponics?

## ASSIGNMENT (WEEK 8)

1. Look around your own home, compound, or school.
2. Identify three (3) "un-used" spaces that could be used for urban horticulture (e.g., a flat roof, a balcony, a patch of bare soil, a sunny wall).
3. For *each* space, suggest one appropriate "practice" (e.g., container garden, vertical garden) and one "crop" (e.g., pepper, lettuce, pumpkin) that could be grown there.

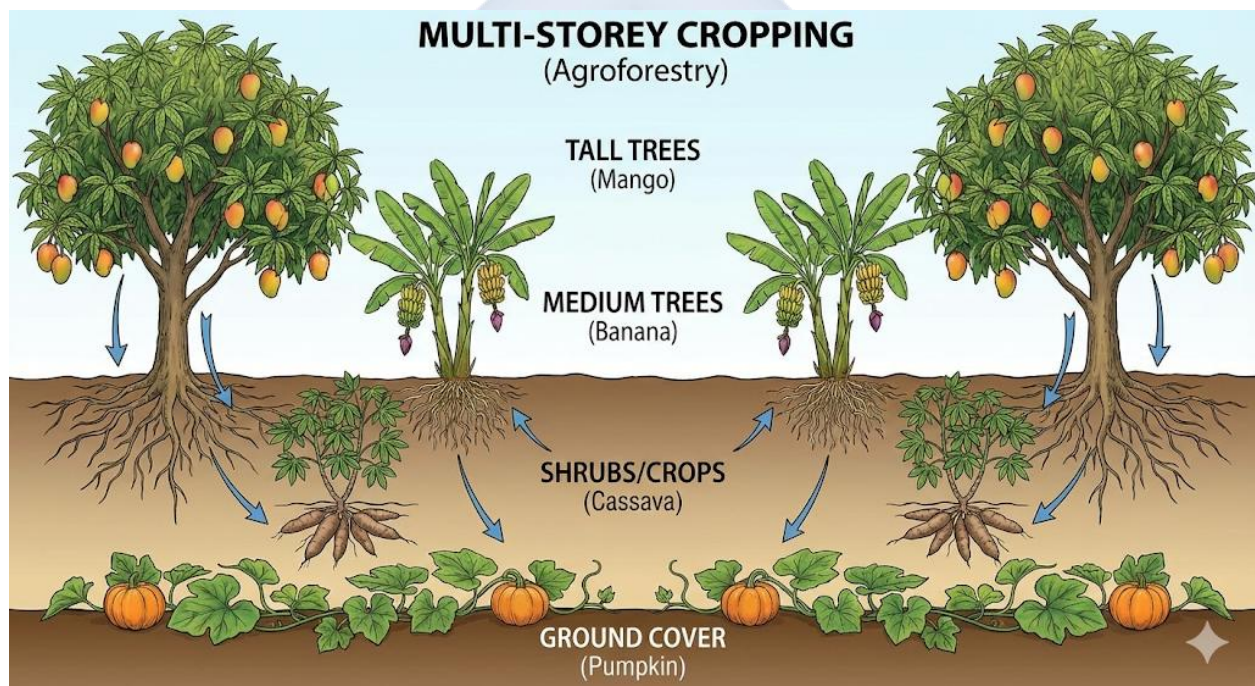


## WEEK 9: AGROFORESTRY

### 9.1 MEANING AND DEFINITION

**Agroforestry** is a sustainable land management system that deliberately *combines* woody perennials (trees, shrubs) with agricultural crops and/or livestock, either at the same time or in sequence, on the same piece of land.

It is the *opposite* of "monoculture" (planting only one crop) and is a more integrated and "natural" way of farming. A traditional Nigerian compound farm (with fruit trees, yams, vegetables, and chickens) is a classic example of agroforestry.



### 9.2 AGROFORESTRY PRACTICES (SYSTEMS)

#### 1. Alley Cropping (Hedgerow Intercropping):

- **What:** Planting food crops (like maize, cassava, or vegetables) in the "alleys" or "lanes" between rows of fast-growing, nitrogen-fixing trees or shrubs (like *Leucaena*, *Gliricidia*, or *Faidherbia albida*).



- **How it works:** The trees are pruned regularly, and the leaves (which are rich in nitrogen) are used as **mulch** and "**green manure**" for the crops in the alleys.
2. **Multi-storey Cropping (Home Gardens):**
- **What:** A system that copies a natural forest, with different "layers" of crops.
  - **Example:**
    - **Top Layer (Tall trees):** Coconut, Oil Palm, Mango, Ogbono.
    - **Middle Layer (Medium trees/shrubs):** Pawpaw, Banana, Plantain, Cocoa, Coffee.
    - **Low Layer (Food crops):** Cassava, Yams (with trellises), Cocoyam, Peppers.
    - **Ground Cover (On the soil):** Pumpkin, Melon, Sweet Potato.
3. **Silvopasture:**
- **What:** Combining trees with *pasture* (grass) and *livestock* (animals).
  - **How it works:** Animals (goats, sheep, cattle) graze on the grass growing *under* the trees. The trees provide shade for the animals and can also be a source of fodder (food).
4. **Windbreaks / Shelterbelts:**
- **What:** One or more rows of trees and shrubs planted around the *border* of a farm or field.
  - **Purpose:** To protect crops, soil, and animals from strong, damaging winds. They also prevent wind erosion.

### 9.3 IMPORTANCE AND BENEFITS OF AGROFORESTRY

1. **Improved Soil Fertility:**

- Leguminous trees (like *Gliricidia*) "fix" nitrogen from the air.
- Tree leaf litter (mulch) decomposes, adding rich organic matter to the soil.
- Tree roots can "pump" nutrients from deep in the soil back up to the surface.

2. **Soil and Water Conservation:**

- Tree roots bind the soil, preventing **water and wind erosion**.
- The "canopy" (tree leaves) slows down heavy rain, so it "soaks in" rather than "runs off."

- Mulch on the soil surface reduces water evaporation.
- 3. **Diversified Income (Economic Security):**
  - The farmer is not reliant on *one* crop.
  - They get short-term income (food crops), medium-term income (fruit), and long-term income (timber, firewood). This spreads the risk.
- 4. **Improved Microclimate:**
  - The tree canopy provides shade, which lowers soil temperature and reduces heat stress on crops and animals.
- 5. **Environmental Benefits:**
  - Increases biodiversity (provides a habitat for birds and "good" insects).
  - Sequesters carbon (trees absorb CO<sub>2</sub>), helping to fight climate change.

## EVALUATION (WEEK 9)

1. Define agroforestry in your own words.
2. What is "Alley Cropping"? Explain how the trees help to fertilize the crops.
3. Sketch and label the four "layers" of a multi-storey home garden.
4. List four (4) distinct benefits of agroforestry.
5. What is the primary function of a "windbreak"?

## ASSIGNMENT (WEEK 9)

1. Read about the tree *Gliricidia sepium* (often called "Quick-stick").
2. Write a short report (one page) on why this tree is so important in agroforestry systems in West Africa. (Hint: Look up its uses for fodder, green manure, and living fences).

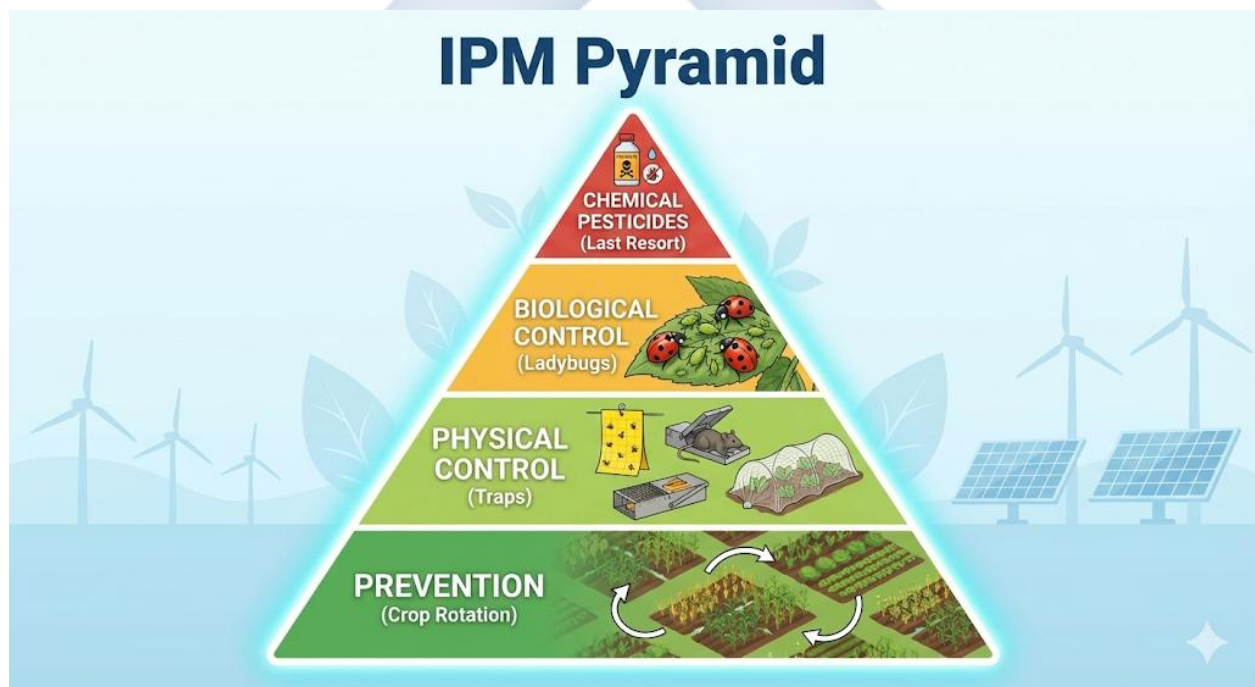
## WEEK 10: ENVIRONMENTAL SUSTAINABILITY IN HORTICULTURE

### 10.1 INTRODUCTION

**Sustainable horticulture** means growing food in a way that is:

1. **Ecologically Sound:** Does not harm the environment (soil, water, air).
2. **Economically Viable:** Farmers can make a good living.
3. **Socially Just:** Good for families and communities.

The goal is to meet our *present* need for food without compromising the ability of *future generations* to meet their own needs. This topic ties together everything we have learned this term (organic farming, agroforestry, water management).



### 10.2 KEY PILLARS OF ENVIRONMENTAL SUSTAINABILITY

#### A. SOIL CONSERVATION AND HEALTH

- **The Problem:** Soil erosion, loss of organic matter, and "dead" soil.
- **Sustainable Practices:**

1. **Conservation Tillage (Zero/Minimum Tillage):** A method of planting *without* ploughing or tilling the soil. A new crop is planted directly into the "stubble" (residue) of the previous crop. This *protects* the soil from erosion, saves water, and builds organic matter.
2. **Cover Cropping & Mulching:** (From Week 4 & 9) The number one rule of soil conservation: **Never leave the soil bare.** A cover crop (like *Mucuna*) or a layer of mulch (like straw) acts as "armour" for the soil, protecting it from sun, wind, and rain.
3. **Contour Ploughing / Terracing:** On *sloping* land, all farming activities (ploughing, planting) *must* be done **across** the slope (on the contour), not *up and down* it. This creates small ridges that trap water and prevent it from "rushing" downhill and causing erosion.

## B. WATER MANAGEMENT AND CONSERVATION

- **The Problem:** Water is scarce, rainfall is unreliable, and traditional irrigation is wasteful.
- **Sustainable Practices:**
  1. **Drip Irrigation:** (From Week 6) The most efficient method. Delivers water in slow "drips" directly to the root zone of each plant. This eliminates evaporation and runoff, saving 70-90% of water.
  2. **Rainwater Harvesting:** Collecting and storing rainwater.
    - *Small-scale:* From roofs into plastic tanks.
    - *Large-scale:* Building small farm ponds or "earth dams" to capture and store runoff water for use in the dry season.
  3. **Mulching:** (As above) A layer of mulch is also one of the best *water conservation* techniques, as it dramatically reduces water loss from evaporation at the soil surface.

## C. INTEGRATED PEST MANAGEMENT (IPM)

- **The Problem:** Over-reliance on chemical pesticides kills beneficial insects, pollutes water, and creates "resistant" pests.

- **IPM is a "middle-ground" approach:** It's a pyramid strategy.
  1. **Base (Foundation): Prevention:** Use *cultural* methods first (crop rotation, resistant varieties, farm sanitation).
  2. **Middle: Biological/Physical:** Use *biological* controls (beneficial insects) and *physical* controls (traps, hand-picking).
  3. **Top (Last Resort): Chemicals:** Use chemical pesticides *only* when the pest level is high enough to cause economic damage, and use the *least-toxic, most-specific* chemical available.

#### D. ENHANCING BIODIVERSITY

- **The Problem:** Monoculture (planting a single crop) is an "invitation" to pests and diseases.
- **Sustainable Practices:**
  - **Polyculture / Intercropping:** Planting many different crops together.
  - **Agroforestry:** (From Week 9) Integrating trees creates habitats.
  - **Planting Flower Borders:** Attracts pollinators (bees) and beneficial insects (pest predators).

#### EVALUATION (WEEK 10)

1. What are the three "pillars" of sustainability? (Ecological, Economic, ...)
2. Define "Conservation Tillage." How does it protect the soil?
3. Explain why farming *on the contour* (across a slope) is a vital soil conservation practice.
4. What is "Rainwater Harvesting"? Give one small-scale and one large-scale example.
5. What is "Integrated Pest Management (IPM)"? At what stage is it acceptable to use a chemical pesticide in an IPM system?

#### ASSIGNMENT (WEEK 10)

1. Draw a diagram of an "IPM Pyramid."
2. Label the four levels, from the *bottom* (most-used) to the *top* (least-used).
  - Level 1: Cultural & Preventive

- Level 2: Physical & Mechanical
- Level 3: Biological
- Level 4: Chemical (Last Resort)





## WEEK 11: REVISION

- **Review of Key Concepts:**

- **Week 1 (Improvement):** Aims (yield, quality, resistance). Methods (Selection, Hybridization - Emasculation, Pollination).
- **Week 2 (Genetics):** Mendel's Laws (Segregation, Independent Assortment). Terms (Gene, Allele, Dominant, Recessive, Phenotype, Genotype). Hybrid Vigour.
- **Week 3 (Biotech):** Tissue Culture (Explant, Sterilization, Callus, Acclimatization, Disease-free). GM Crops (Bt, Herbicide Tolerance).
- **Week 4 (Organic):** Principles (Health, Ecology). Practices (Compost, Green Manure, Crop Rotation, Neem Oil, Mulching).
- **Week 6 (Greenhouse):** Types (Glass, Polyunnel, Net House). Benefits (Year-round, Protection, Water-saving). Challenges (Cost, *Overheating*).
- **Week 8 (Urban):** Importance (Food security, Income). Practices (Containers, Sack gardens, Vertical). Hydroponics (NFT, DWC, Pros/Cons).
- **Week 9 (Agroforestry):** Systems (Alley cropping, Multi-storey, Silvopasture, Windbreaks). Benefits (Soil fertility, Conservation, Diversified income).
- **Week 10 (Sustainability):** Soil (Contour ploughing, Zero-tillage). Water (Drip irrigation, Rain harvesting). Pest (IPM Pyramid).

## EVALUATION (WEEK 11)

1. What is the difference between Mendel's Law of Segregation and the Law of Independent Assortment?
2. A farmer has a virus-infected banana plant but wants to propagate it. What *biotechnology* technique could be used to create clean, virus-free plantlets?
3. Compare "Alley Cropping" and "Hydroponics" based on their (a) use of soil, (b) use of technology, and (c) key benefits.
4. A farmer with a field on a gentle slope is losing topsoil to erosion. Recommend three (3) *sustainable practices* to help solve this problem.

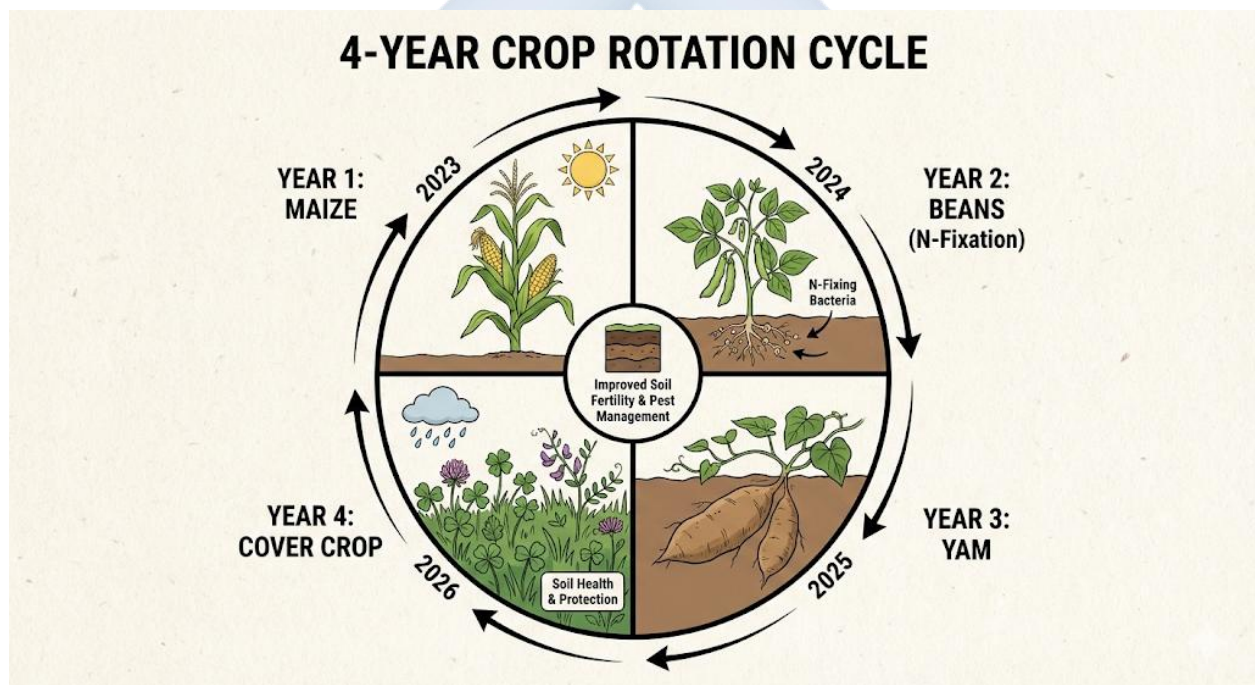
## SS2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

### SECOND TERM

#### WEEK 1: CROP ROTATION

##### 1.1 INTRODUCTION

Crop rotation is a fundamental and ancient agricultural practice. It is a systematic, planned sequence of growing different types of crops (from different plant families) on the same piece of land over a period of years. It is the opposite of **monocropping** (growing the same crop on the same land year after year), which is a practice that can degrade soil and increase pest problems.



##### 1.2 PRINCIPLES OF CROP ROTATION

The success of crop rotation relies on several key agronomic principles. The plan is designed to ensure that different crops, with their unique characteristics, contribute to the soil and the farm ecosystem in a balanced way.

###### 1. Nutrient Management (Legumes vs. Feeders):

- **Heavy Feeders:** Some crops, like maize and tomatoes, are "heavy feeders" that consume large amounts of soil nutrients, especially nitrogen.
- **Nitrogen-Fixers (Legumes):** Crops from the legume family (e.g., cowpea, groundnuts, soybeans, *Mucuna*) have a symbiotic relationship with bacteria (*Rhizobium*) in their root nodules. These bacteria "fix" atmospheric nitrogen (N<sub>2</sub>) into a form (ammonia) that plants can use.
- **The Principle:** By planting a legume crop *after* a heavy feeder, the farmer naturally replenishes the soil's nitrogen supply, reducing the need for expensive synthetic fertilizers.

## 2. Pest and Disease Cycle Disruption:

- Many soil-borne pests (like nematodes) and diseases (like fungal wilts) are **crop-specific**. They build up in the soil when their preferred host crop is planted repeatedly.
- **The Principle:** When you plant a non-host crop, the pest or pathogen "starves" or its life cycle is broken. For example, the nematodes that attack yams will not survive on a crop of maize. By rotating, the pest population crashes, ensuring a healthier field for the next time the susceptible crop is planted.

## 3. Varying Rooting Depths (Soil Structure):

- **Shallow-rooted crops** (e.g., onions, lettuce) primarily use the top layer of soil.
- **Deep-rooted crops** (e.g., cassava, sunflowers, some cover crops) can break through compacted soil layers ("hardpan") and draw nutrients and water from deep within the soil profile.
- **The Principle:** Rotating between deep and shallow-rooted crops improves soil aeration, water infiltration, and overall soil structure. The deep-rooted crops also "recycle" nutrients by bringing them up to the surface, where they become available to the next crop after the plant decomposes.

## 4. Weed Control:

- Different crops compete with weeds differently.
- **Smother Crops:** Some crops (like cassava, sweet potatoes, or *Mucuna*) develop a dense canopy that "chokes" or "smothers" weeds by blocking sunlight.
- **Poor Competitors:** Other crops (like onions or carrots) are easily overgrown.

- **The Principle:** Rotating a "smother crop" into the sequence can naturally reduce the "weed seed bank" in the soil, making it easier to manage weeds in the next, more sensitive crop.

### 1.3 ADVANTAGES OF CROP ROTATION

1. **Improved Soil Fertility & Health:** Balances nutrient levels, increases soil organic matter (when cover crops are used), and promotes a healthy soil microbiome.
2. **Reduced Chemical Use:** Decreases the need for synthetic fertilizers (due to legumes) and pesticides/nematicides (due to pest cycle disruption). This saves the farmer money and protects the environment.
3. **Increased Crop Yields:** Healthier, more fertile soil leads to higher yields over the long term.
4. **Risk Diversification:** By planting different crops, the farmer is less vulnerable. If market prices for one crop crash, or a specific pest attacks, they still have other crops to rely on for income.
5. **Better Soil Structure:** Improves water holding capacity and aeration, and reduces soil erosion.

### 1.4 EXAMPLES OF A 4-YEAR CROP ROTATION PLAN

A good rotation plan will typically include a crop from each of these categories:

- **A - Fruit/Leaf:** (e.g., Maize, Tomato, Okra) - *Heavy Feeder*
- **B - Legume:** (e.g., Cowpea, Groundnut, Soybean) - *Nitrogen Fixer*
- **C - Root/Tuber:** (e.g., Yam, Cassava, Potato) - *Heavy Potash (K) Feeder*
- **D - Fallow/Cover Crop:** (e.g., *Mucuna*, *Pueraria*) - *Soil Builder*

#### Example Plan for a Single Plot of Land:

- **Year 1: Maize** (Heavy nitrogen feeder)
- **Year 2: Cowpea** (Replenishes nitrogen)

- **Year 3: Yams** (Use the high organic matter and residual nitrogen; followed by cassava, which is a deep-rooted, heavy potash feeder)
- **Year 4: Fallow with *Mucuna*** (To rest the soil, add massive organic matter, and control weeds)

### EVALUATION (WEEK 1)

1. Define Crop Rotation.
2. What is the main difference between crop rotation and monocropping?
3. Explain the principle of "Pest and Disease Cycle Disruption."
4. Why is it a good idea to plant a legume (like groundnuts) *after* a cereal (like maize)?
5. Design a simple 3-year crop rotation for a local farm, and justify your choice for each year.

### ASSIGNMENT (WEEK 1)

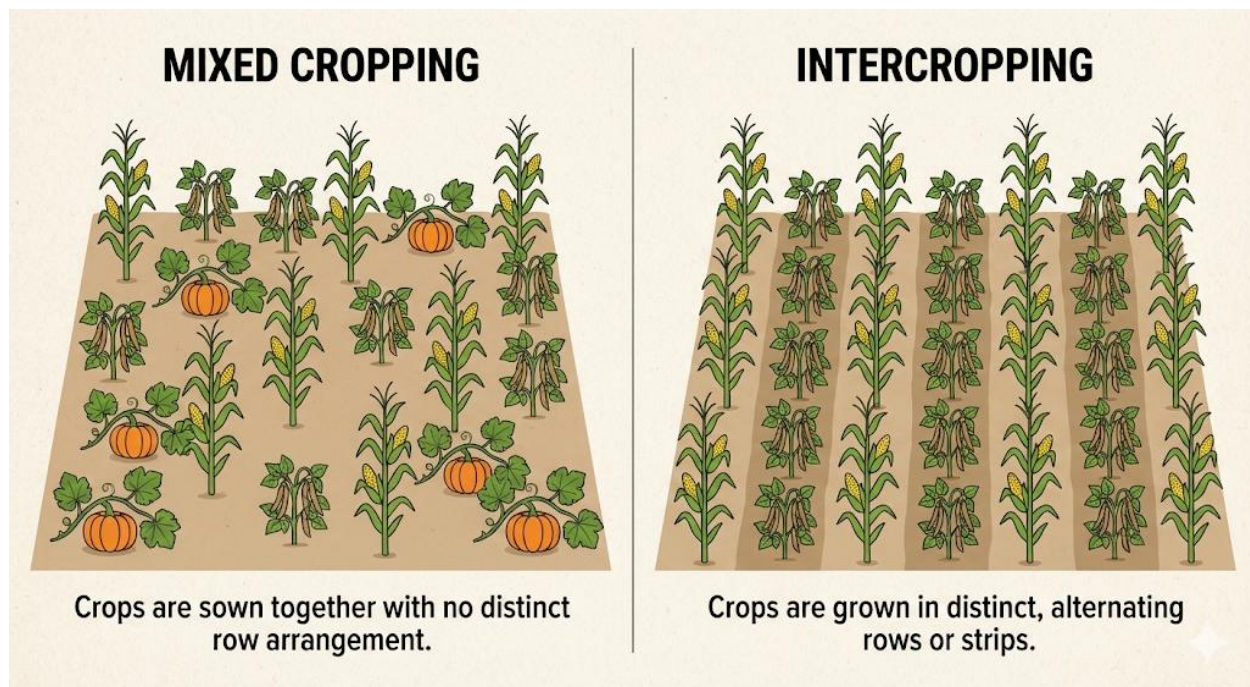
1. Interview a local farmer (a family member or neighbour).
2. Ask them if they practice crop rotation. If yes, what is their sequence of crops?
3. If they practice monocropping, what challenges do they face (e.g., "The soil is tired," "I have to buy more fertilizer," "Weevils are a big problem")?
4. Write a one-page report on your findings.



## WEEK 2: MIXED CROPPING AND INTERCROPPING

### 2.1 INTRODUCTION

These are traditional farming systems, common throughout Nigeria and West Africa, where multiple crops are grown on the same piece of land at the same time. This is a form of **polyculture** and contrasts sharply with the "monoculture" of large-scale commercial farming.



### 2.2 DEFINITIONS

- **Mixed Cropping (or Mixed-Intercropping):**
  - **Definition:** The practice of growing two or more crops *simultaneously* on the same field, with **no distinct row arrangement**.
  - **Example:** A traditional farm where maize, cassava, and pumpkin seeds are all planted, scattered or mixed together in the same space. It appears random.
- **Intercropping (or Strip-Cropping/Row Intercropping):**
  - **Definition:** The practice of growing two or more crops *simultaneously* on the same field in a **specific, patterned arrangement**. This is a more systematic and modern approach.



- **Examples:**

- **Row Intercropping:** One row of maize, followed by one row of cowpea, followed by one row of maize.
- **Strip Intercropping:** A 1-meter "strip" of maize, followed by a 1-meter "strip" of soybeans.

## 2.3 BENEFITS OF MIXED/INTERCROPPING

1. **Risk Diversification (Main Benefit):** This is the farmer's "insurance policy." If one crop fails due to drought, pests, or low prices, the other crops may survive and provide food and income. It is the principle of "not putting all your eggs in one basket."
2. **Higher Total Yield (Land Equivalent Ratio - LER):** LER is a measure of efficiency. An LER greater than 1.0 means the intercrop is more productive than growing the crops separately. For example, if you get 0.7-ha-worth of maize and 0.5-ha-worth of cowpea from a 1-ha intercrop, your LER is 1.2. This happens because the crops use resources (light, water, nutrients) more efficiently together.
3. **Pest and Disease Suppression:**
  - **Pest Confusion:** Pests that find their host plant by smell are "confused" by the other crop's smell.
  - **Trap Cropping:** One crop (the "trap crop") can be planted to attract pests *away* from the main, more valuable crop.
  - **Physical Barrier:** Tall crops (like maize) can act as a barrier, preventing flying pests from reaching the shorter crop (like beans).
4. **Weed Suppression:** Planting low-growing, spreading crops (like melon, pumpkin, or sweet potato) as a "ground cover" under a tall crop (like maize or cassava) smothers weeds by blocking sunlight.
5. **Improved Soil Fertility:** When a legume (like cowpea) is intercropped with a cereal (like maize), the legume provides some nitrogen to the cereal.
6. **Better Use of Resources:** Different crops have different needs:
  - **Light:** A tall, sun-loving crop (maize) can grow alongside a short, shade-tolerant crop (cocoyam).

- **Roots:** A deep-rooted crop (cassava) can grow with a shallow-rooted crop (onions), using different soil layers.

## 2.4 LIMITATIONS (CHALLENGES)

1. **Mechanization is Difficult or Impossible:** This is the *biggest* disadvantage. You cannot use tractors, planters, or combine harvesters on a mixed-crop field. This limits the farm size and relies heavily on manual labour.
2. **Interspecific Competition:** If the crops are not chosen carefully, they will *compete* with each other for light, water, and nutrients. This can lead to *lower* yields for both crops.
3. **Complex Management:** All operations (planting, fertilizing, weeding, harvesting) are more complicated. The crops may have different planting times, nutrient needs, and harvesting times.
4. **Pest Control Complexity:** A pesticide that is safe for one crop might kill the other (e.g., a herbicide for maize might kill the intercropped beans).

## EVALUATION (WEEK 2)

1. What is the main difference between Mixed Cropping and Intercropping?
2. Explain the concept of "Risk Diversification" in the context of mixed cropping.
3. What is the "Land Equivalent Ratio (LER)"? What does an LER of 1.3 mean?
4. Give one example of how intercropping can help control weeds.
5. What is the single greatest limitation of mixed/intercropping that prevents it from being used on large commercial farms?

## ASSIGNMENT (WEEK 2)

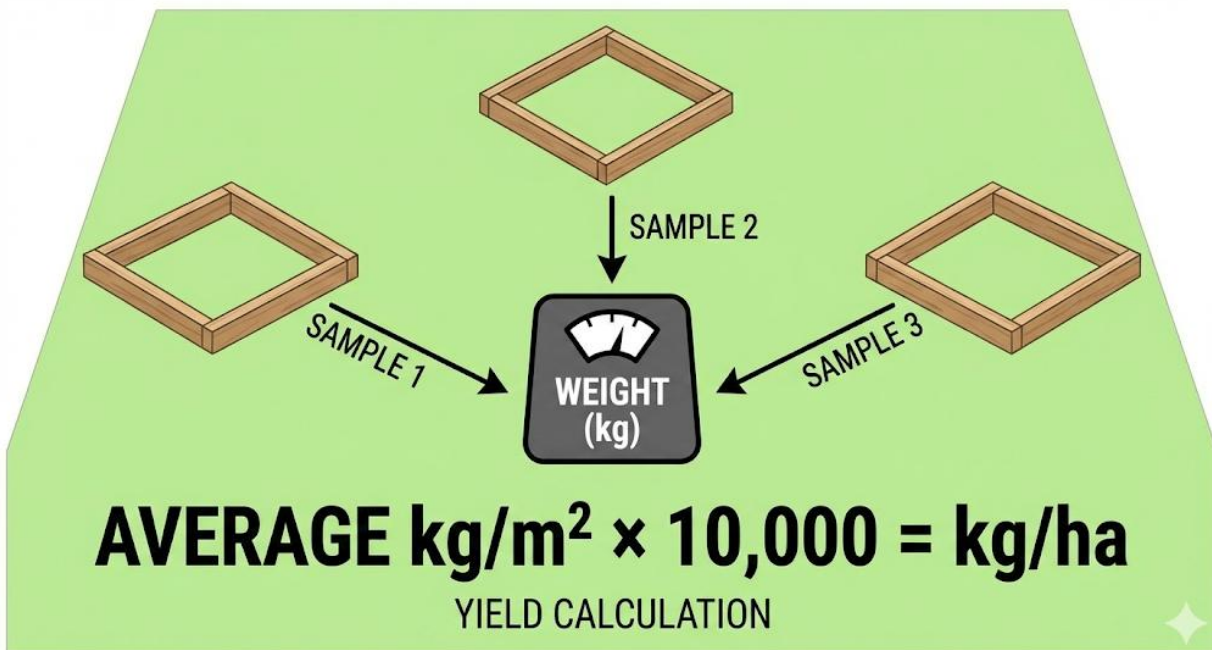
1. Visit a local farm or market garden (or observe farms on the side of the road).
2. Identify a field where intercropping or mixed cropping is being practiced.
3. Draw a simple diagram of the field. Identify the crops being grown together.
4. Write a short paragraph explaining the *likely benefits* the farmer is getting from this specific combination (e.g., "The farmer is growing maize and melon. The melon (ground cover) is suppressing weeds and the maize (cereal) is providing food").

## WEEK 3: CROP YIELD MEASUREMENT

### 3.1 INTRODUCTION

"Yield" is the measure of agricultural output (the harvested product) per unit area of land. It is the most common way to measure farm productivity and efficiency.

- **Definition:** Yield is the mass, volume, or count of a crop produced from a defined unit of land.
- **Standard Unit:** The standard international unit for yield is **kilograms per hectare (kg/ha)** or **tonnes per hectare (t/ha)**.



### 3.2 IMPORTANCE OF MEASURING YIELD

Why bother measuring yield accurately?

1. **Economic Performance:** To calculate profit and loss. A farmer must know their yield (income) to compare against their costs (inputs).
2. **Decision Making & Comparison:**

- **Variety Trials:** To know which variety of maize (e.g., 'A' vs. 'B') *actually* produces more.
- **Treatment Effects:** To scientifically test if a new fertilizer or a new irrigation method *actually* increased production.
- **Farm Management:** To identify which fields are high-performing and which are low-performing.

### 3. Planning and Forecasting:

- **Farmer:** To know how much storage to prepare and how much income to expect.
- **Government/NGO:** To forecast national food production, predict potential famines, and manage food imports/exports.

## 3.3 METHODS OF CROP YIELD MEASUREMENT

### 1. FULL FIELD MEASUREMENT (CENSUS METHOD)

- **Process:** The farmer harvests the *entire* field. The *entire* harvest is collected, cleaned, and weighed. The total weight is then divided by the total area of the field.
- **Example:** A farmer harvests a 2-hectare field and gets a total of 8,000 kg of dry maize.
  - Yield = 8,000 kg / 2 ha = 4,000 kg/ha (or 4 t/ha).
- **Pros:** 100% accurate.
- **Cons:** Only practical for the farmer *after* harvest; not useful for researchers who need to sample many plots; time-consuming.

### 2. SAMPLING METHOD (QUADRANT ANALYSIS)

This is the standard scientific and estimation method used by researchers, agricultural officers, and assessors.

- **Process:**
  1. **Define the Sampling Unit:** A small, fixed-size frame (a "quadrant") is used. A common size is **1 meter x 1 meter (1m<sup>2</sup>)**.

2. **Randomization:** The quadrant is thrown or placed *randomly* at multiple locations in the field. This is critical to avoid bias (e.g., not just picking the "best-looking" spots).
  3. **Harvest and Weigh:** The crop *inside* the quadrant is carefully harvested, and its weight is recorded.
  4. **Repeat:** This is done multiple times (e.g., 10-20 times) across the field to get a reliable average.
  5. **Calculate the Average:** All the weights are added up and divided by the number of samples.
    - e.g., 10 samples (throws) produce a total of 6 kg.
    - Average = 6 kg / 10 samples = 0.6 kg per 1m<sup>2</sup> (the average yield).
- **Extrapolation (The Calculation):**
    - **Step 1:** You know the average yield per 1m<sup>2</sup> (e.g., 0.6 kg/m<sup>2</sup>).
    - **Step 2:** You know the number of square meters in one hectare: **1 ha = 10,000 m<sup>2</sup>**.
    - **Step 3:** Multiply the average yield by the area.
      - Estimated Yield = 0.6 kg/m<sup>2</sup> × 10,000 m<sup>2</sup> = **6,000 kg/ha** (or 6 t/ha).

### 3.4 IMPORTANT CONSIDERATIONS

1. **Moisture Content:** The yield of grain (like maize) must be corrected to a standard moisture content (e.g., 12.5% or 15%). A farmer harvesting wet maize (25% moisture) will have a higher weight, but it's not the "true" dry yield.
2. **Edge Effects:** The outer rows of a field often behave differently (more sun, more wind). Samples should avoid the very edge of the field.
3. **Harvestable Yield vs. Biological Yield:** "Biological yield" is the total plant mass. "Harvestable (or Economic) yield" is *only* the part that is sold (e.g., the grain, not the stalk).

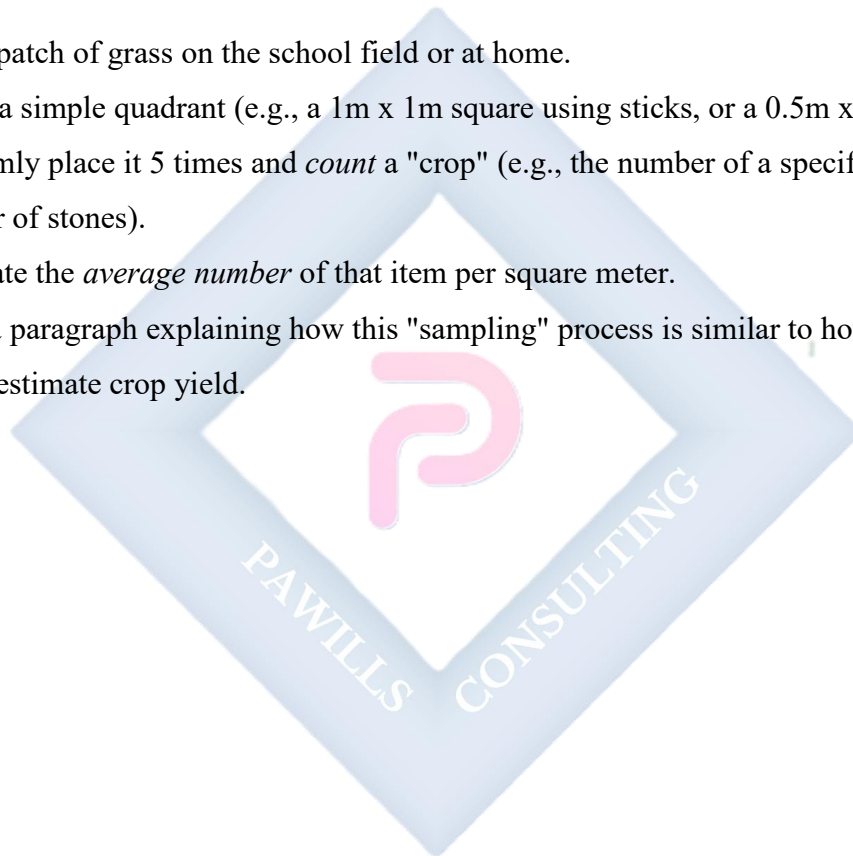
### EVALUATION (WEEK 3)

1. Define "Crop Yield" and state the standard international unit used to measure it.
2. List three (3) distinct reasons why it is important to measure yield.

3. Describe the "Full Field Measurement" (Census) method.
4. Explain the process of the "Sampling Method" (Quadrant Analysis).
5. You use a  $1\text{m}^2$  quadrant to sample a maize field. You take 5 samples with the following weights: 0.5 kg, 0.4 kg, 0.6 kg, 0.5 kg, 0.5 kg.
  - (a) What is the *average* yield in  $\text{kg}/\text{m}^2$ ?
  - (b) What is the *extrapolated* yield in  $\text{kg}/\text{ha}$ ?

### ASSIGNMENT (WEEK 3)

1. Find a patch of grass on the school field or at home.
2. Create a simple quadrant (e.g., a  $1\text{m} \times 1\text{m}$  square using sticks, or a  $0.5\text{m} \times 0.5\text{m}$  square).
3. Randomly place it 5 times and *count* a "crop" (e.g., the number of a specific weed, or the number of stones).
4. Calculate the *average number* of that item per square meter.
5. Write a paragraph explaining how this "sampling" process is similar to how a scientist would estimate crop yield.

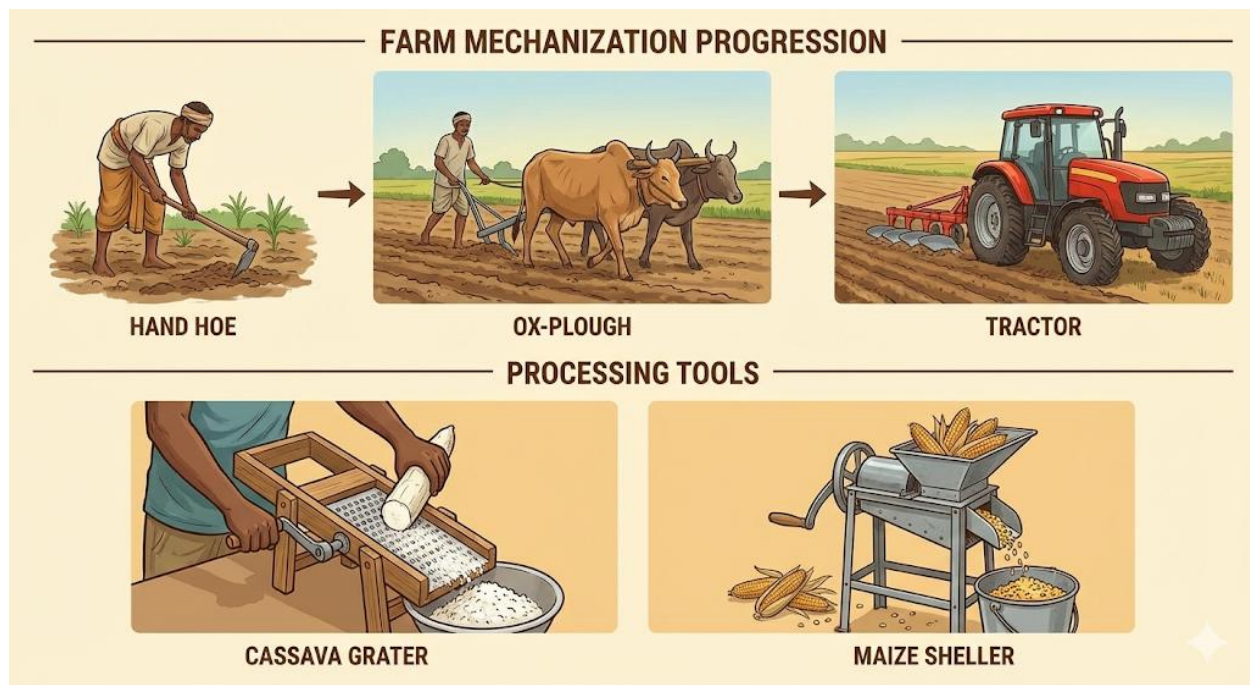




## WEEK 4: FARM MECHANIZATION IN CROP PRODUCTION

### 4.1 INTRODUCTION

Farm mechanization is the process of using agricultural machinery to replace human and animal (draught) labour in farming operations. It ranges from simple hand tools to advanced GPS-guided combine harvesters. It is a key component of modernizing agriculture.



### 4.2 LEVELS OF MECHANIZATION

#### 1. Level 1: Hand-Tool Technology (Manual):

- The farmer uses simple tools (hoe, cutlass, sickle, digging stick).
- *Pros*: Low cost, no maintenance, suitable for any land size.
- *Cons*: Extremely slow, high drudgery (back-breaking work), limits farm size to 1-2 ha (subsistence).

#### 2. Level 2: Animal-Draught Technology:

- Uses trained animals (e.g., oxen, donkeys) to pull simple implements (plough, ridger, cart).
- *Pros*: Faster than manual, can cultivate larger areas (5-10 ha), provides manure.

- *Cons:* Animals get sick, need food/water, requires training.

### 3. **Level 3: Engine-Powered Technology (Motorized):**

- Uses engines (petrol or diesel) to power machinery.
- *Examples:* Tractors, combine harvesters, water pumps, motorized tillers, maize shellers.

## 4.3 SIMPLE MACHINES & EQUIPMENT (The focus for smallholders)

While tractors are a symbol of mechanization, *intermediate* mechanization is more practical for many farmers.

- **Land Preparation:** Hand-held tillers, small 2-wheel tractors (power tillers).
- **Planting:** Hand-push planters, simple seed drills.
- **Weed Control:** Motorized brush cutters.
- **Harvesting:** Small-scale rice reapers.
- **Post-Harvest Processing (Crucial):**
  - **Cassava Graters:** Motorized graters have revolutionized garri production.
  - **Rice Threshers/Mills:** Separate grain from stalk and husk.
  - **Maize Shellers:** Quickly remove kernels from the cob.
  - **Oil Palm Presses:** Extract palm oil much more efficiently than by hand.

## 4.4 ADVANTAGES OF MECHANIZATION

1. **Timeliness of Operations:** This is the *most important* advantage. In rain-fed agriculture, a farmer has a very small "window" to plant. A tractor can plough 5 hectares in one day; a man with a hoe would take weeks.
2. **Reduces Drudgery:** Frees farmers from the hardest, most boring, and back-breaking physical labour.
3. **Increases Farm Size (Scalability):** Mechanization allows a single farmer to cultivate a much larger area of land, moving from subsistence to commercial farming.
4. **Improved Efficiency and Quality:**
  - *Planting:* A planter places seeds at a uniform depth and spacing, leading to better germination.

- *Harvesting*: A combine harvester reduces grain loss.
- *Processing*: A cassava grater or oil press extracts more product with less waste.

#### 4.5 LIMITATIONS AND CHALLENGES IN NIGERIA

1. **High Initial Cost**: This is the #1 barrier. Tractors and implements are extremely expensive, far beyond the reach of most small-scale farmers.
2. **Lack of Maintenance & Spare Parts**: A tractor is useless if it's broken. There is a severe shortage of trained mechanics and affordable, genuine spare parts in rural areas.
3. **Land Tenure & Fragmentation**: Most farmers have small (0.5-2 ha) plots that are scattered in different locations. This "land fragmentation" is highly *unsuitable* for large machinery.
4. **Soil Compaction**: Heavy tractors, especially in wet soil, can compact the earth, creating a "hardpan" that roots cannot penetrate and water cannot infiltrate.
5. **Job Displacement**: Mechanization can lead to rural unemployment, as one machine can do the work of 50 labourers.
6. **Lack of Technical Know-How**: Farmers need to be trained to operate and maintain the machinery correctly.

#### EVALUATION (WEEK 4)

1. Define Farm Mechanization.
2. What is the *most important* advantage of mechanization, especially in rain-fed agriculture?
3. Give two (2) examples of simple, post-harvest processing machines.
4. List four (4) major limitations or challenges to farm mechanization in Nigeria.
5. What is "Land Fragmentation," and why is it a barrier to mechanization?

#### ASSIGNMENT (WEEK 4)

1. Choose *one* common post-harvest operation (e.g., shelling maize, grating cassava, pressing palm oil).
2. Describe the "traditional/manual" method of doing this job.
3. Describe the "mechanized" method of doing the same job.

4. List three advantages the mechanized method has over the manual one.

## **WEEK 5: MIDTERM EXAMINATION**

*This week is for assessment of all topics covered in Weeks 1-4.*



## WEEK 6: IRRIGATION SYSTEMS

### 6.1 INTRODUCTION

Irrigation is the artificial application of water to land to supplement rainfall. It is essential for:

- **Dry Season Farming:** Growing crops year-round, not just in the rainy season.
- **Drought Mitigation:** Saving a crop during an unexpected "dry spell."
- **Arid/Semi-Arid Regions:** Making agriculture possible in dry areas.

### 6.2 TRADITIONAL IRRIGATION (THE PROBLEM)

- **Surface / Flood Irrigation:** This involves "flooding" the entire field or running water down "furrows" (channels) between crop rows.
- **Pros:** Very cheap to set up, low-tech, requires no special equipment.
- **Cons: Extremely wasteful.** Up to 60% of the water is lost to evaporation, deep percolation (sinking too deep), and runoff. It also promotes weed growth and can lead to soil salinization.



### 6.3 MODERN METHODS (WATER CONSERVATION)



Modern systems focus on "Water Use Efficiency" (WUE) – getting "more crop per drop."

## 1. SPRINKLER IRRIGATION

- **How it Works:** Water is pumped through pipes and sprayed into the air through "sprinkler heads" (e.g., impact sprinklers, pop-up sprinklers, or large "rain guns"). It falls on the crops like artificial rain.
- **Pros:**
  - Much more efficient (approx. 75-85%) than flood irrigation.
  - Suitable for most crops and soil types.
  - Can cover large areas.
- **Cons:**
  - High initial cost (pumps, pipes, sprinklers).
  - High energy cost to pressurize the water.
  - Water can be lost to wind drift and evaporation *before* it even hits the ground.
  - Wets the leaves, which can increase fungal diseases.
- **Types:** Center-Pivot (giant, moving arms for massive fields), Rain Guns (for large, open fields).





## 2. DRIP IRRIGATION (MICRO-IRRIGATION)

- **How it Works:** Water is delivered *very slowly*, in *drops* or a "micro-stream," *directly* to the soil at the base (root zone) of *each individual plant*. It uses a network of plastic pipes ("drip lines") and small nozzles called "emitters."



- **Pros:**
  - **Highest Efficiency:** The most water-efficient method (90-98%).
  - **Water Conservation:** Virtually *no* water is lost to evaporation or runoff.
  - **Weed Control:** The "alleys" between rows stay dry, which suppresses weed growth.
  - **Fertigation:** Allows for "fertigation" – the application of liquid fertilizer *through* the irrigation water, delivering nutrients directly to the roots.
  - Reduces fungal diseases (as leaves stay dry).
- **Cons:**
  - **High Initial Cost:** Very expensive to install (pipes, emitters, filters).
  - **Clogging:** The tiny emitters can easily get clogged by sand, dirt, or algae.  
*Requires filters and maintenance.*

- **Not for all crops:** Best for high-value, row crops (vegetables, orchards, greenhouses), not for broadcast crops like wheat.

## EVALUATION (WEEK 6)

1. Define Irrigation.
2. What is the main *problem* (disadvantage) of "Surface/Flood Irrigation"?
3. Describe how "Sprinkler Irrigation" works.
4. Describe how "Drip Irrigation" works.
5. You are given a small plot to farm high-value tomatoes in a dry region. You have a limited water supply. Which *modern* irrigation system would you choose, and *why*? (List 3 reasons for your choice).

## ASSIGNMENT (WEEK 6)

1. Draw a simple diagram comparing Sprinkler Irrigation and Drip Irrigation.
2. Your diagram for Drip Irrigation should show the water going to the *roots* of individual plants.
3. Your diagram for Sprinkler Irrigation should show the water being *sprayed* over the whole area.
4. Label the key parts (pump, pipes, sprinkler head, drip emitter).

## **WEEK 7: MIDTERM BREAK**

*This is a break from instruction. Students are encouraged to review the first half of the term's work.*



## WEEK 8: STORAGE STRUCTURES

### 8.1 INTRODUCTION

**Post-harvest losses** are a massive problem in Nigeria. It is estimated that 30-50% of food (especially perishables) is lost *after* it is harvested, before it reaches a consumer. Proper storage is the key to preventing this loss.



#### Importance of Storage:

1. **Prevent Post-Harvest Losses:** Protects crops from spoilage, pests (weevils, rats), and moisture.
2. **Ensure Food Security:** Allows food to be available during the "lean season" (off-season).
3. **Price Stabilization (Economic):** Allows a farmer to *store* their harvest when prices are low (at harvest time) and *sell* it months later when prices are high.

### 8.2 TYPES OF STORAGE STRUCTURES

#### 1. FOR DRY GRAINS (MAIZE, RICE, COWPEA, SOYBEANS)



- **Traditional (Low-tech):**
  - **Rhombus:** A traditional mud-walled, thatched-roof structure, often raised on stilts. *Problem:* Prone to rodents and insects.
  - **Bag Storage:** Storing in woven polypropylene (WPP) or jute bags, stacked on wooden "pallets" (not on the bare floor) in a ventilated room.
- **Modern (Improved):**
  - **Silos:** Large, cylindrical towers made of metal or concrete. Used for *bulk* (unbagged) storage of grain on a large commercial scale (e.t., by big farms or food processing companies like Flour Mills). They are airtight and protect from all pests and weather.
  - **Hermetic Storage Bags (e.g., PICS Bags):** A "low-tech, high-impact" modern solution. It's a "bag-within-a-bag" system. An inner plastic liner is made of a special material that is **airtight**. After dry grain is put in and the bag is sealed, the insects (weevils) inside use up the oxygen, *suffocate themselves*, and die.

## 2. FOR TUBERS (YAMS, POTATOES)

- **Traditional:**
  - **Yam Barn:** A traditional open-air structure with vertical wooden poles and cross-pieces. Yams are tied individually to the frame. It is designed for maximum **ventilation** to prevent rot and allow farmers to monitor the yams.
  - **Sand Boxes:** Storing root crops (like carrots) in boxes of cool, slightly damp sand.
- **Modern:**
  - **Ventilated Sheds:** Modern "barns" made of wood/metal, with slatted shelves to store tubers in a cool, dark, and well-ventilated space.

## 3. FOR PERISHABLES (TOMATOES, PEPPERS, FRUITS, VEGETABLES)

- **The Problem:** These have high water content and "breathe" (respire) fast, so they spoil very quickly. They need *cold* storage.
- **Refrigerators:** For small-scale, short-term storage (e.g., retailers, homes).
- **Cold Rooms (Walk-in Coolers):**

- **Definition:** A large, insulated room or container with a refrigeration unit. It is a walk-in refrigerator.
- **Function:** They are the *backbone* of the modern food supply chain. They slow down respiration, inhibit microbial growth, and can extend the shelf life of tomatoes from 3 days to 21 days.
- **Evaporative Coolers:** A low-cost, "off-grid" alternative. A simple frame covered with charcoal or jute bags, which are kept wet. As water evaporates, it draws heat from the air inside, making the "cooler" 10-15°C colder than the outside air.

### EVALUATION (WEEK 8)

1. What is a "post-harvest loss"? Give two (2) main reasons *why* farmers should store their crops.
2. What is a "Silo," and what type of crop is it used for?
3. Describe how a "Hermetic Bag" (like a PICS bag) controls insects.
4. Why is "ventilation" the most important principle for a traditional "Yam Barn"?
5. What is the main difference between a "Silo" and a "Cold Room" in terms of what they store and how they work?

### ASSIGNMENT (WEEK 8)

1. Design a simple, low-cost "storage structure" for *one* of the following crops: (a) Yams, (b) Maize (in bags), (c) Tomatoes (short-term).
2. Draw your design.
3. Label the key features and write a short paragraph explaining *why* your design will help prevent spoilage (e.g., "I used pallets to keep the bags off the floor," "I used netting for ventilation," "I used charcoal and water for cooling").



## WEEK 9: AGRICULTURAL EXTENSION SERVICES

### 9.1 INTRODUCTION

There is a large "knowledge gap" in agriculture. Researchers at universities (e.g., OAU, ABU) and institutes (e.g., IITA, NCRI) develop new technologies (new seeds, new methods, new pest controls). However, most farmers, especially in rural areas, do not have access to this information.

- **Definition:** An **Agricultural Extension Service** is a system that "bridges" this knowledge gap. The **Extension Agent (or Officer)** is the *link*, the *messenger*, and the *teacher* who takes knowledge from the *researcher* and extends it *to the farmer*. It is also a two-way street, as they take problems *from the farmer* back *to the researcher*.



### 9.2 ROLES OF AGRICULTURAL EXTENSION

The role of the extension agent is multi-faceted:

1. **Education and Training:** This is the primary role. They teach farmers *modern* and *appropriate* farming practices (e.g., how to use fertilizer, correct crop spacing, pest identification, crop rotation).
2. **Technology Transfer:** They are responsible for introducing new technologies to farmers, such as:
  - Improved seed varieties (e.g., disease-resistant cassava).
  - Simple machines (e.g., maize shellers).
  - New methods (e.g., drip irrigation).
3. **Facilitation and Farmer Organization:** They help farmers form groups, such as the **cooperatives** (see Week 10). This helps farmers get a collective voice, access loans, and buy inputs in bulk.
4. **Feedback Channel:** The extension agent listens to the farmers' problems (e.g., "A new insect is destroying our maize," "The new cassava variety yields well, but the garri does not taste good"). They then report this *feedback* to the researchers, who can then try to solve the *real-world* problem.
5. **Advisory Services:** Acting as a "farm doctor," visiting farms, diagnosing problems (pests, diseases, nutrient deficiencies), and recommending solutions.

### 9.3 EXTENSION TEACHING METHODS

- **Individual Methods:**
  - **Farm & Home Visits:** Visiting a farmer directly. Very effective, but slow (can only see a few farmers a day).
- **Group Methods:**
  - **Demonstration Plots (Method & Result):** This is the *most effective* method.
    - The agent sets up two small plots *side-by-side*.
    - **Plot A (Farmer's Practice):** Uses the traditional method.
    - **Plot B (Recommended Practice):** Uses the new seed/fertilizer.
    - At the end of the season, all the farmers *see* the result. "Seeing is believing."
  - **Farmers' Field Days:** A large meeting at the demonstration plot to discuss the results.

- **Mass Media Methods:**

- **Radio:** The most important mass-media tool in rural Africa. Farmers listen to agricultural programs in their local language.
- **Posters, Leaflets, and (increasingly) Mobile Phones (SMS/Video).**

## 9.4 IMPORTANCE OF EXTENSION

- Improves farm productivity and farmer income.
- Increases national food security.
- Accelerates the adoption of new, helpful technologies.
- Connects researchers to the real-world problems of farmers.
- Empowers rural communities.

## EVALUATION (WEEK 9)

1. Define "Agricultural Extension." What is the "knowledge gap" it tries to bridge?
2. What is the name of the person who acts as the "link" between the researcher and the farmer?
3. List four (4) distinct roles of the extension service.
4. Explain what a "Demonstration Plot" is and why it is such an effective teaching method.
5. Why is "radio" considered such an important tool for mass-media extension?

## ASSIGNMENT (WEEK 9)

1. Imagine you are an extension agent, and you need to teach farmers in your village how to use "Drip Irrigation."
2. Which *teaching method* (Individual, Group, or Mass Media) would you use, and *why*?
3. Describe the steps you would take to teach this new technology.

## WEEK 10: AGRICULTURAL COOPERATIVES

### 10.1 INTRODUCTION

**The Core Problem:** A single, small-scale farmer has *no bargaining power*.

- When they **buy inputs** (fertilizer, seeds), they buy in small quantities at a *high* retail price.
- When they **sell produce** (maize, cocoa), they sell small quantities to a "middleman" at a *low* farm-gate price.

**Definition:** An **Agricultural Cooperative Society** is a business organization that is *voluntarily owned, controlled, and operated* by its *members* (the farmers) for their mutual social and economic benefit. It is a "self-help" group.



### 10.2 FUNCTIONS (TYPES) OF COOPERATIVES

Cooperatives solve the "core problem" by using the power of *bulk buying and selling*.

#### 1. Marketing Cooperatives:



- **Function:** Farmers *pool* or *bulk* their produce together. Instead of 100 farmers selling 1 bag of cocoa each, the cooperative sells 100 bags (or more) *directly* to a large buyer or exporter.
- **Benefit:** They *cut out the middleman* and negotiate a much *higher price* for all their members.

## 2. Purchasing (Input Supply) Cooperatives:

- **Function:** The cooperative buys fertilizer, seeds, and chemicals *in bulk* (wholesale) directly from the manufacturer.
- **Benefit:** It then sells these inputs to its members at a *low price*, much cheaper than the local retail shop.

## 3. Credit Cooperatives (Savings & Loans):

- **Function:** Members contribute regular savings (e.g., ₦500 per week). This pool of money ("capital") is then given out as small, low-interest loans to members.
- **Benefit:** Provides farmers with access to credit to buy inputs or hire labour, which traditional banks will not provide.

## 4. Service Cooperatives:

- **Function:** The cooperative *jointly owns* or *hires* a large piece of equipment that no single farmer could afford.
- **Benefit:** Provides members with access to services like a *tractor*, a *storage warehouse (silo)*, or a *processing machine* (e.g., a "Cooperative Rice Mill").

# 10.3 ADVANTAGES OF COOPERATIVES

1. **Economic:** Higher profits for farmers (sell high, buy low).
2. **Bargaining Power:** Gives farmers a "collective voice" in the market and in dealing with the government.
3. **Access to Credit & Services:** Provides access to loans, inputs, and machinery.
4. **Empowerment:** Trains farmers in leadership, business management, and record keeping.
5. **Risk Reduction:** Helps members manage risks and supports them in times of need.

# 10.4 CHALLENGES OF COOPERATIVES IN NIGERIA

1. **Mismanagement and Corruption:** This is the #1 cause of failure. Poor leadership or outright theft of funds destroys the cooperative.
2. **Low Member Participation:** If members do not attend meetings or feel involved, they lose trust and the cooperative weakens.
3. **Lack of Capital:** Many cooperatives struggle to raise enough capital to buy a tractor or build a warehouse.
4. **Poor Loan Repayment:** Members taking loans and defaulting (refusing to pay back) can bankrupt a credit cooperative.

### EVALUATION (WEEK 10)

1. Define "Agricultural Cooperative." Who *owns* and *controls* it?
2. What is the "core problem" a single, small-scale farmer faces that a cooperative tries to solve?
3. Explain the function of a "Marketing Cooperative."
4. Explain the function of a "Purchasing (Input Supply) Cooperative."
5. List two (2) major advantages and two (2) major challenges of cooperatives.

### ASSIGNMENT (WEEK 10)

1. You and 50 other students in your class are all small-scale poultry (chicken) farmers. You are all facing high costs for feed.
2. Write a one-page proposal on *why* you should all form a "Cooperative."
3. Explain *what type* of cooperative it would be and *what function* it would perform to solve your "high feed cost" problem.



## WEEK 11: REVISION

- **Review of Key Concepts:**

- **Week 1 (Crop Rotation):** Principles (Nutrients, Pests, Soil, Weeds) & Advantages (Fertility, Less chemicals, Risk reduction).
- **Week 2 (Intercropping):** Definitions (Mixed vs. Intercropping) & Benefits (Risk, LER, Pest/Weed suppression) & Limitations (Mechanization).
- **Week 3 (Yield):** Importance (Economics, Decisions) & Methods (Census vs. Sampling) & Calculation (kg/ha).
- **Week 4 (Mechanization):** Advantages (Timeliness, Scale) & Limitations (Cost, Maintenance, Land Fragmentation).
- **Week 6 (Irrigation):** Problem (Flood) & Modern Methods (Sprinkler vs. Drip) & Benefit of Drip (Efficiency, Fertigation, Weed control).
- **Week 8 (Storage):** Importance (Post-harvest loss, Price) & Structures (Silos, Barns, Cold Rooms) & Low-cost Tech (PICS bags).
- **Week 9 (Extension):** Definition (The "Link") & Roles (Education, Tech Transfer, Feedback) & Methods (Demo Plot).
- **Week 10 (Cooperatives):** Definition (Member-owned) & Functions (Marketing, Purchasing, Credit, Service) & Benefits (Bargaining power).

## EVALUATION (WEEK 11)

1. What is the main difference between Crop Rotation (Week 1) and Intercropping (Week 2)?
2. A farmer with 20 hectares of land wants to grow maize. What is the *main barrier* (Week 4) to him using a tractor, and *why* would an Extension Agent (Week 9) be needed to help him?
3. A tomato farmer wants to start dry-season farming. What *irrigation system* (Week 6) would you recommend, and what *storage structure* (Week 8) would he need to invest in to market his produce effectively?
4. How can a Cooperative (Week 10) help a group of farmers solve the problem of high fertilizer costs *and* low selling prices?

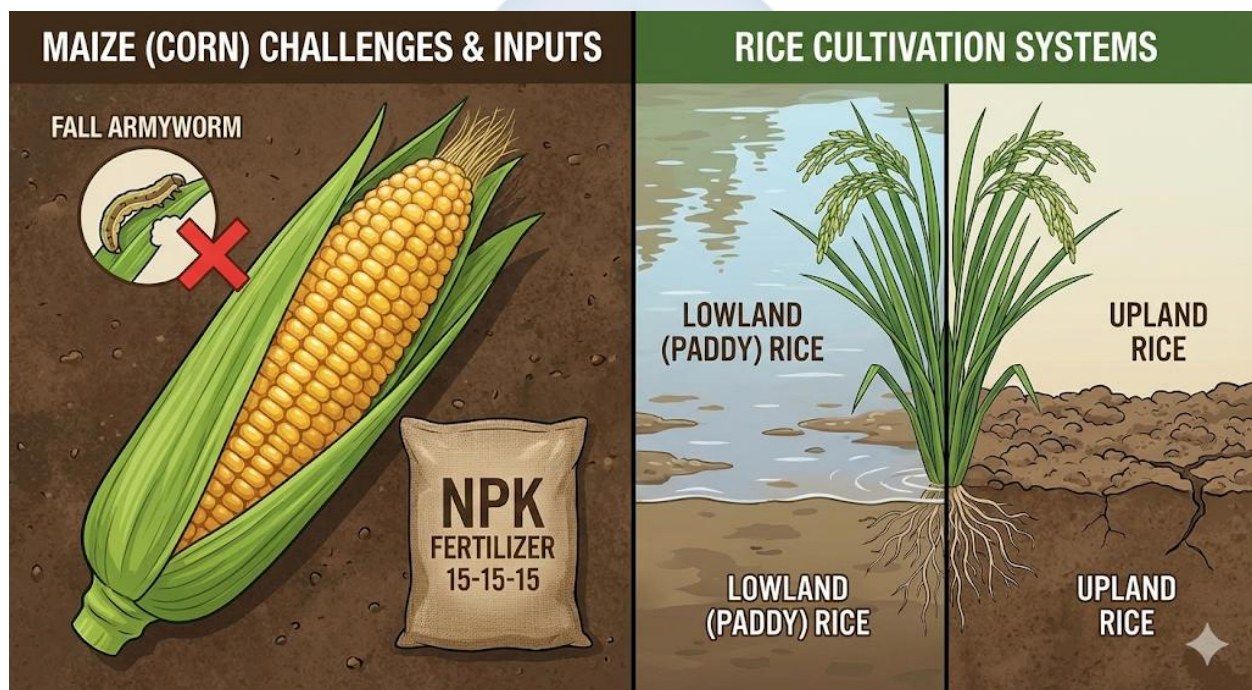
## SS2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

### THIRD TERM

#### WEEK 1: CEREAL CROPS

##### 1.1 INTRODUCTION TO CEREALS

Cereal crops (or grains) are plants from the grass family (Poaceae) cultivated for their edible starchy seeds (grains). They are the most important source of carbohydrates and staple food for the majority of the world's population. In Nigeria, they are critical for food security.



##### 1.2 MAIZE (*Zea mays*)

- **Importance:** A primary staple food (eaten as 'agbado', 'oka', pap, or 'eko'), a key component of animal feed (especially for poultry), and an industrial raw material (for flour, corn starch, oil).
- **Ecology:**

- **Climate:** Requires warm, sunny weather. Needs adequate rainfall (600-1500mm) spread throughout the growing season, especially during the "tasseling" (flowering) and grain-filling stages. It needs a dry period for harvest and drying.
- **Soil:** Prefers deep, well-drained, fertile loamy soils with a high organic matter content. pH 6.0-7.0.
- **Cultivation Practices:**
  - **Land Preparation:** Clear land, plough, and harrow to create a fine tilth. Ridges or beds can be made.
  - **Planting:** Planted by seed. In Nigeria, there are two seasons: *Early* (March-April) and *Late* (August-September).
  - **Spacing:** 75-90cm between rows, 25-30cm within rows. Plant 2-3 seeds per hole, 3-5cm deep.
  - **Fertilizer:** Maize is a *heavy feeder*, especially of Nitrogen. A split application is best:
    1. **At Planting:** Apply NPK 15:15:15 or 20:10:10 (basal application).
    2. **4-6 Weeks After Planting:** Top-dress with Urea (Nitrogen) to boost vegetative growth.
  - **Weed Control:** This is *critical* in the first 6 weeks. Weeds severely reduce yield. Can be done by hoeing or with pre-emergence (e.g., Atrazine) and post-emergence herbicides.
- **Pests & Diseases:**
  - **Pests:** Stem Borers (larvae tunnel into the stem), Armyworms (especially the **Fall Armyworm**, which devastates crops), Weevils (in storage).
  - **Diseases:** Maize Streak Virus, Rust, Downy Mildew.
- **Harvesting:** 90-120 days after planting. Matures when the husk turns brown and dry, and the "black layer" forms at the kernel base.
- **Post-Harvest:** Harvest cobs, de-husk, dry (to 12-14% moisture), shell (remove kernels), and store (in bags, PICS bags, or silos) to protect from weevils.

### 1.3 RICE (*Oryza sativa*)

- **Importance:** A major staple food in Nigeria, with enormous (and growing) demand.

- **Ecology (Two Main Types):**

1. **Lowland/Swamp/Paddy Rice:** Grown in flooded fields ("paddies") or river valleys (fadamas). Requires bunds (walls) to hold water.
2. **Upland Rice:** Grown on rain-fed land, just like maize or sorghum.

- **Cultivation Practices:**

- **Land Preparation:**

- **Lowland:** Plough, harrow, and then "puddle" the soil (mix soil and water) to create a soft, level, water-holding mud.
    - **Upland:** Plough and harrow to a fine tilth.

- **Planting:**

1. **Direct Seeding:** Broadcasting or drilling seeds directly into the field.
2. **Transplanting:** (Common for lowland) Seeds are first grown in a "nursery" for 21-30 days. The seedlings are then uprooted and transplanted manually into the puddled field.

- **Water Management:** This is the key. Lowland rice needs 5-10cm of standing water for most of its growth. Upland rice relies entirely on rainfall.
  - **Weed Control:** *The* major problem, especially in upland rice. Flooding in lowland rice naturally suppresses many weeds.

- **Pests & Diseases:**

- **Pests:** Stem Borers, **Birds** (a major pest that eats ripening grain), Termites.
  - **Diseases:** Rice Blast, Rice Yellow Mottle Virus (RYMV).

- **Harvesting:** 100-150 days. Harvest when 80-90% of the panicles (heads) are golden-yellow and drooping. Cut with sickles.

- **Post-Harvest:** **Thresh** (beat to separate grain from stalk), **Dry** (on tarps or floors), **Winnow** (remove chaff), **Parboil** (a key process in Nigeria: soak, steam, and re-dry to improve milling and nutrition), **Mill** (remove husk and bran), **De-stone**.

#### 1.4 SORGHUM (*Sorghum bicolor*) & MILLET (*Pennisetum glaucum*)

- **Importance:** *The* primary staple food crops of the arid and semi-arid Northern zones (Sudan and Sahel savanna). They are vital for food security in dry regions.

- **Ecology:** Their defining feature is **drought tolerance**.
  - **Climate:** Thrive in high temperatures and low rainfall (300-600mm) where maize and rice would fail.
  - **Soil:** Can grow on poorer, sandier soils.
- **Cultivation:** Similar land preparation to maize, but planted on ridges. Often intercropped with cowpea.
- **Pests & Diseases:**
  - **Pests:** Stem Borers, **Birds (Quelea birds)**, which fly in massive flocks and can destroy a whole field.
  - **Diseases:** Smut (replaces grain with a black powder), Downy Mildew.
- **Harvesting:** Cut the seed heads (panicles), dry them, and thresh.

### EVALUATION (WEEK 1)

1. Why is maize considered a "heavy feeder," and what is the standard 2-step fertilizer application method for it?
2. What is the *most significant* ecological difference between Upland Rice and Lowland Rice?
3. What is "Parboiling" in rice processing?
4. Why are sorghum and millet the dominant cereals in Northern Nigeria?
5. Name one major pest of Maize, one of Rice, and one of Sorghum.

### ASSIGNMENT (WEEK 1)

1. Define "Fall Armyworm." Find out when it arrived in Nigeria and why it is such a devastating pest for maize farmers. Write a one-page report.



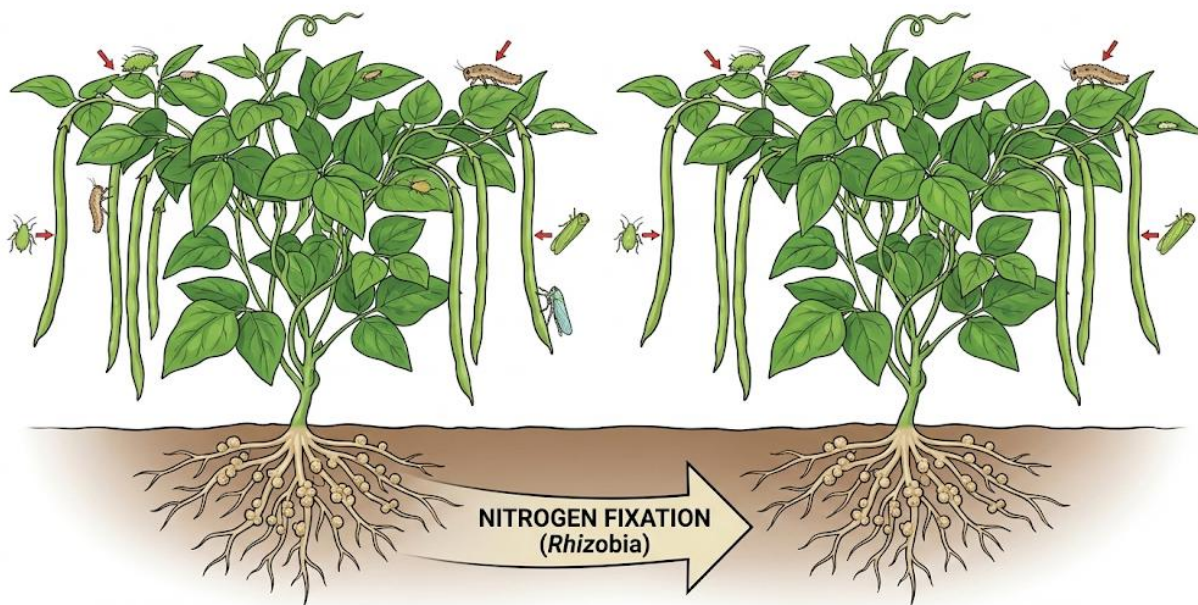
## WEEK 2: LEGUMINOUS CROPS

### 2.1 INTRODUCTION TO LEGUMES

Legumes (from the Fabaceae family) are crops grown for their edible seeds ("pulses") or pods.

- **Key Characteristic:** Their ability to **fix nitrogen**. They have nodules on their roots containing *Rhizobium* bacteria, which convert atmospheric nitrogen ( $N_2$ ) into a usable form.
- **Importance:**
  1. **Protein:** They are the "poor man's meat," a critical source of plant-based protein.
  2. **Soil Fertility:** They naturally enrich the soil with nitrogen, making them perfect for crop rotation and intercropping with cereals.

### LEGUMINOUS CROPS: COWPEA



### 2.2 COWPEA (*Vigna unguiculata*)

- **Importance:** The most important food legume in Nigeria (used for beans, 'akara', 'moin-moin').

- **Ecology:** A warm-weather crop, relatively drought-tolerant (but needs water at flowering). Prefers well-drained sandy loam.
- **Cultivation:**
  - **Planting:** Plant seeds directly, 2-3 per hole. Spacing (e.g., 60x30cm).
  - **Fertilizer:** *Does not* require Nitrogen fertilizer. A small "starter" dose of NPK or just P and K (e.g., SSP) is sufficient.
- **Pests & Diseases:**
  - **Pests:** This is the **major limiting factor** for cowpea. It is attacked by a *complex* of pests at every stage:
    - **Seedling:** Aphids.
    - **Flowering:** Thrips (cause flowers to dry up and fall off).
    - **Podding:** Maruca Pod Borer (larva bores into pods).
    - **Late Pod:** Pod-sucking Bugs (cause shrivelled seeds).
  - **Control:** Requires multiple, timed insecticide sprays to get a good yield.
- **Harvest & Storage:** Pods are picked as they dry, then threshed. The grains must be protected from **Bruchids (Weevils)**, which destroy them in storage. **PICS bags** (hermetic storage) are an excellent solution.

### 2.3 SOYBEAN (*Glycine max*)

- **Importance:** A major global commodity. Used for oil (vegetable oil), animal feed (Soybean Meal), and food (soymilk, *awara* - local tofu).
- **Ecology:** Needs *more* water and *better* soil than cowpea.
- **Cultivation:**
  - **Inoculation:** In new fields, seeds *must* be inoculated with the correct bacteria (*Bradyrhizobium japonicum*) to ensure nitrogen fixation.
  - **Planting:** Drilled in rows. Weeds must be controlled.
- **Harvesting:** A key challenge is **pod shattering**. The whole plant is harvested (cut or uprooted) when 90% of leaves have fallen and pods are dry. If left too long, pods will shatter and spill the seeds.
- **Processing:** Threshed, dried, and stored.

## 2.4 GROUNDNUT / PEANUT (*Arachis hypogaea*)

- **Importance:** Grown for its edible seeds (eaten raw, boiled, roasted) and for its oil.
- **Botany (Geocarpy):** A unique plant. It flowers above ground, but after pollination, a "peg" (stalk) grows from the flower *down into the soil*, where the pod (the "nut") develops.
- **Ecology:** Requires *light, sandy, friable (loose) soil* to allow the pegs to penetrate easily.
- **Cultivation:** Planted on ridges or beds.
- **Pests & Diseases:**
  - **Groundnut Rosette Disease:** A virus (spread by aphids) that stunts the plant.
  - **Aflatoxin:** A *major* problem. A fungus (*Aspergillus flavus*) grows on the pods if they are not dried properly. This fungus produces aflatoxin, a deadly toxin and potent carcinogen.
- **Harvesting:** Uproot the entire plant. **Curing** (letting the plants dry in the field for a few days) is essential to reduce moisture. Pods are then picked and dried thoroughly.

### EVALUATION (WEEK 2)

1. What is the *single most important* agronomic (soil) benefit of planting legumes?
2. List the three (3) main pests that attack cowpea *after* flowering.
3. Why are "PICS bags" recommended for storing cowpea?
4. What is "Geocarpy" in groundnuts?
5. What is Aflatoxin, and under what conditions does it develop?

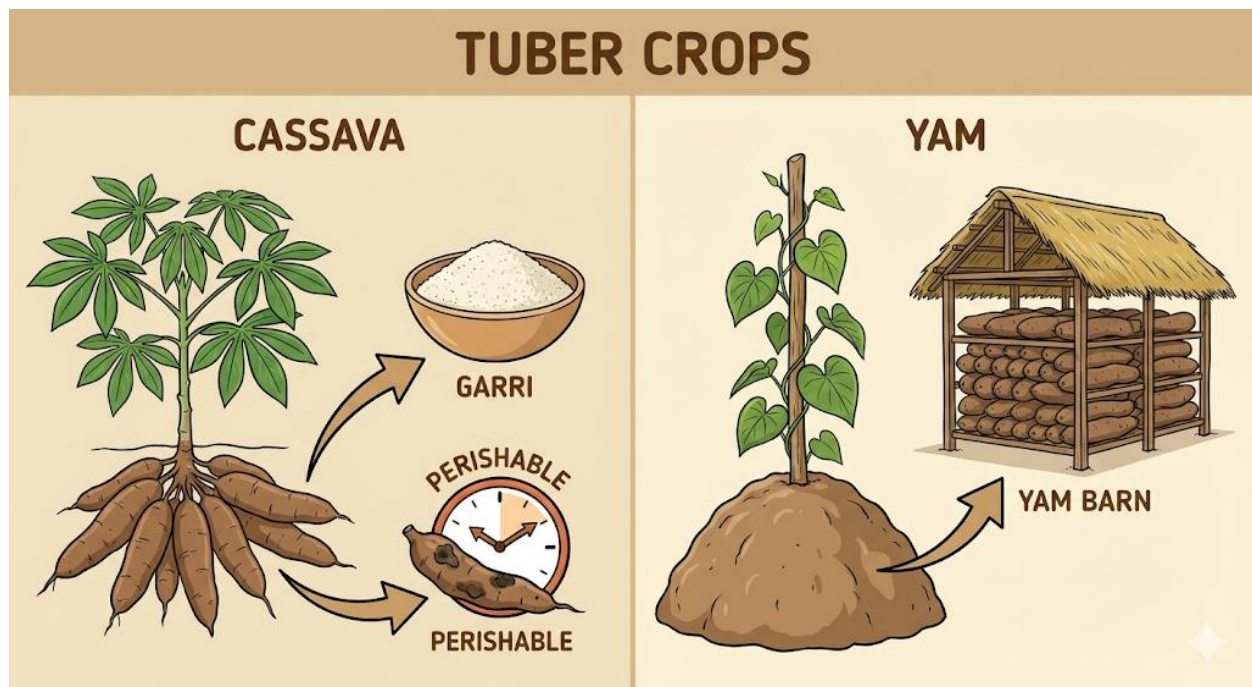
### ASSIGNMENT (WEEK 2)

1. Draw a simple diagram of a root nodule on a legume.
2. Explain the "symbiotic relationship" between the *Rhizobium* bacteria and the plant (i.e., what does the plant get, and what does the bacteria get?).

## WEEK 3: TUBER CROPS

### 3.1 INTRODUCTION TO TUBERS

Root and Tuber crops are plants grown for their starchy, bulky, underground storage organs. They are staple "energy foods" and crucial for food security in West Africa. They are mostly propagated *vegetatively* (clonally).



### 3.2 CASSAVA (*Manihot esculenta*)

- **Importance:** The most widely cultivated tuber in Nigeria. A food security crop.
- **Ecology:** Its main advantage is its **hardiness**. It tolerates drought, grows on poor/acidic soils where other crops fail, and can be "stored" in the ground (left unharvested).
- **Planting Material:** Stem cuttings (25-30cm long), taken from the woody part of the stem.
- **Cultivation:**
  - **Land Prep:** Planted on *ridges* or mounds to promote tuber growth.
  - **Planting:** Cuttings are planted at an angle, 1m x 1m spacing.
- **The Cyanide Problem:** Cassava roots contain **cyanogenic glycosides** (which produce *cyanide*, a poison).

- **Sweet Varieties:** Low cyanide, can be boiled and eaten.
- **Bitter Varieties:** High cyanide, are *toxic* if eaten raw. Must be processed.
- **Processing (Essential):** Processing (grating, fermenting, boiling, frying) *removes* the cyanide.
  - **Garri:** Peeled, grated, fermented, de-watered, and fried (toasted).
  - **Fufu/Akpu:** Fermented in water, sieved, boiled, and pounded.
  - **Lafun:** Peeled, chopped, dried, and milled into flour.
- **Pests & Diseases:** Cassava Mosaic Disease (CMD) (virus), Cassava Green Mite, Cassava Mealybug.
- **Harvest:** 8-18 months. Uproot roots. **Problem:** Harvested roots are highly *perishable* and will rot in 2-3 days. Must be processed immediately.

### 3.3 YAM (*Dioscorea spp.*)

- **Importance:** A culturally significant crop ("The King of Crops"), especially in Nigeria (the world's largest producer).
- **Ecology:** Requires deep, fertile, loose, well-drained soil. Cannot tolerate waterlogging. Needs high rainfall.
- **Planting Material:** "Setts" (pieces of a yam tuber, 250-500g).
- **Cultivation:**
  - **Land Prep:** Requires *deep* soil. Planted in large **mounds** or high ridges.
  - **Staking:** This is **essential** for most varieties. A stake (pole) is provided for the yam vine to climb. This exposes leaves to maximum sunlight, leading to large tubers.
- **Harvest:** 9-12 months. Requires *careful* digging to avoid bruising the tubers.
- **Storage:** Unlike cassava, yams *can* be stored fresh for months. They are stored in a **Yam Barn** – a shaded, well-ventilated structure where yams are tied to poles.

### 3.4 COCOYAM (*Colocasia esculenta* & *Xanthosoma sagittifolium*)

- **Importance:** A key staple in some parts of the SE/SW.



- **Ecology:** *Opposite* of most crops. It is **shade-tolerant** and **water-loving**. It thrives in damp, humid conditions and is often intercropped *under* tree crops (e.g., cocoa, plantain).
- **Planting Material:** Cormels or small setts.

### 3.5 POTATO (IRISH POTATO) (*Solanum tuberosum*)

- **Ecology:** This is a **temperate crop**, not a tropical one. It requires **cool temperatures** to form tubers.
- **In Nigeria:** It is *only* grown in high-altitude, cool areas: **Jos Plateau (Plateau State)** and **Mambilla Plateau (Taraba State)**.

### EVALUATION (WEEK 3)

1. What is the *main ecological advantage* of cassava?
2. Why is processing (e.g., into garri) *essential* for bitter cassava varieties?
3. Why must cassava be processed within 2-3 days of harvesting?
4. What is "staking" in yam cultivation, and why is it necessary?
5. Why can't "Irish" potatoes be grown in Lagos or Port Harcourt?

### ASSIGNMENT (WEEK 3)

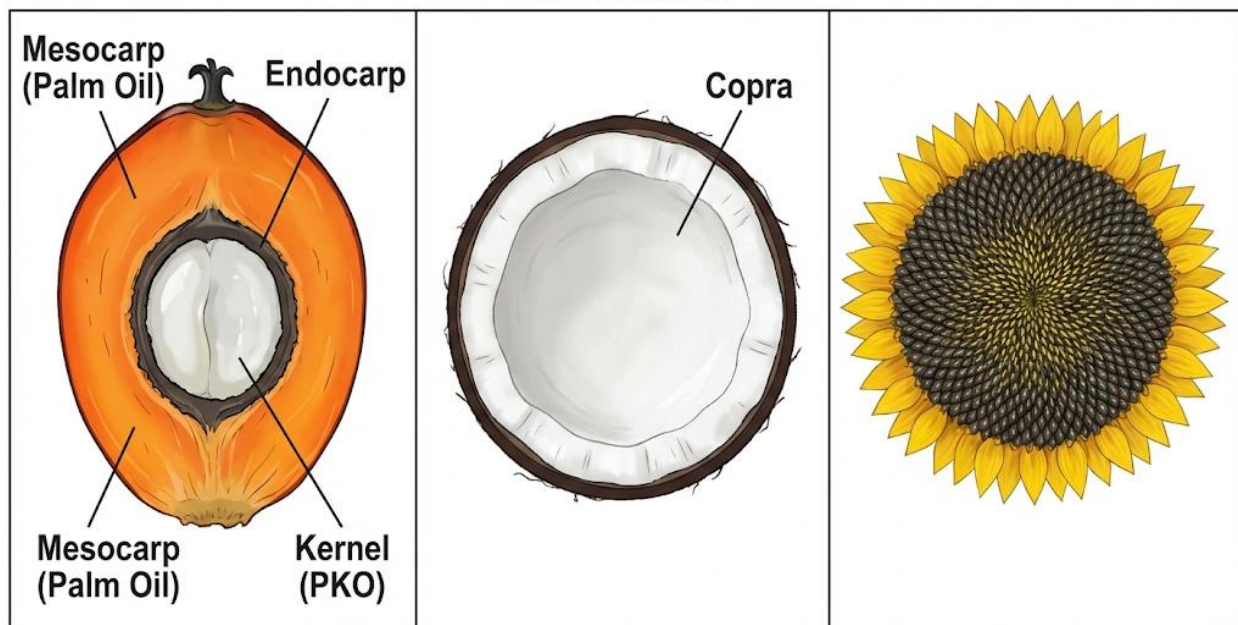
1. Draw a diagram of a traditional "Yam Barn."
2. Write a short paragraph explaining the three key principles of yam storage: (1) Good ventilation, (2) No contact with the ground, (3) Shaded from sun.

## WEEK 4: OIL CROPS

### 4.1 INTRODUCTION

Oil crops are plants cultivated primarily for the oil extracted from their seeds or fruits. These oils are used for cooking, industrial applications, soap making (saponification), and biofuels.

### OIL CROPS



### 4.2 OIL PALM (*Elaeis guineensis*)

- **Importance:** The most important oil crop in Nigeria.
- **Ecology:** A humid tropical crop. Requires high rainfall (2000mm+), high humidity, and high temperatures. Needs deep, moist, fertile soils. (Dominant in the South-East and South-South).
- **Planting Material:** Seedlings. The **Tenera** hybrid (a cross between *Dura* and *Pisifera*) is the preferred commercial variety because it has a thick, fleshy part (mesocarp) and a thin shell, giving the highest oil yield.
- **Cultivation:** A plantation crop. Spacing is wide (e.g., 9m triangular).
- **Harvesting:** Begins 3-4 years after planting. **Fresh Fruit Bunches (FFB)** are harvested using a cutlass (for short trees) or a "Malaysian knife" (a long pole with a sickle).

- **Processing (Two Oils):**

1. **Palm Oil (Red Oil):** Extracted from the outer **fleshy part (mesocarp)**.
  - *Process:* Bunches are sterilized (boiled), fruits are stripped, digested (pounded), **pressed** (to extract oil), and the oil is clarified.
2. **Palm Kernel Oil (PKO):** Extracted from the **nut/kernel** inside the hard shell.
  - *Process:* The nuts are separated, cracked, the kernels are roasted, and then **pressed** to extract PKO.

#### 4.3 COCONUT (*Cocos nucifera*)

- **Importance:** A versatile crop ("tree of life").
- **Ecology:** A coastal crop. It is **salt-tolerant** and thrives in sandy, well-drained soils in high-humidity areas (e.g., Lagos, Badagry).
- **Products:**
  - **Copra:** The dried kernel (flesh). This is the part *from which coconut oil is pressed*.
  - **Coconut Water:** The liquid inside the young nut.
  - **Coir:** The fibre from the outer husk, used for ropes, mats, etc.
  - **Coconut Milk:** Extracted from the grated fresh kernel.

#### 4.4 SUNFLOWER (*Helianthus annuus*)

- **Importance:** A minor but growing oil crop.
- **Botany:** Unlike the others (which are perennial trees), this is an **annual crop** (grows and dies in one season).
- **Ecology:** More adaptable and drought-tolerant than oil palm.
- **Cultivation:** Grown in rows, similar to maize. The large "head" contains the seeds.
- **Products:** High-quality cooking oil and seeds (snacks).

#### EVALUATION (WEEK 4)

1. Name the *two* different types of oil that are extracted from the Oil Palm fruit.
2. Which *part* of the fruit does Palm Oil come from, and which *part* does Palm Kernel Oil come from?

3. Why is the "Tenera" hybrid the preferred variety for oil palm plantations?
4. What is "Copra"?
5. What is the main ecological difference between oil palm and sunflower cultivation?

#### **ASSIGNMENT (WEEK 4)**

1. Draw a cross-section of an Oil Palm fruit.
2. Label the four main parts: (1) Exocarp (skin), (2) Mesocarp (flesh - source of palm oil), (3) Endocarp (shell), (4) Kernel (source of PKO).



## **WEEK 5: MIDTERM EXAMINATION**

*This week is for assessment of all topics covered in Weeks 1-4.*





## WEEK 6: VEGETABLES

### 6.1 INTRODUCTION

Vegetables are parts of plants consumed by humans as food. They are "**protective foods**," essential for health as they provide vitamins, minerals, and fibre. They are high-value, perishable crops.

### TOMATO PRODUCTION TIMELINE



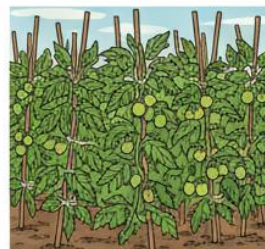
1. Nursery



Whitefly +  
Virus



2. Transplant



3. Staking



4. Dry Season  
Harvest



## 6.2 TOMATO (*Solanum lycopersicum*)

- **Importance:** A major vegetable, used in virtually every Nigerian stew, sauce, and Jollof rice.
- **Ecology:** Requires warm, sunny days. **Problem:** High humidity and rainfall promote fungal diseases. Therefore, **dry season production** (using irrigation) is the most reliable and productive, resulting in better quality fruit.
- **Cultivation:**
  - **Nursery:** Seeds are *not* planted directly. They are raised in a "nursery" (a protected bed) for 3-4 weeks.
  - **Transplanting:** Seedlings are carefully moved to the main field.
  - **Staking:** Indeterminate (tall/climbing) varieties must be staked to keep fruit off the ground, prevent rot, and make spraying easier.
- **Pests & Diseases:**
  - **Diseases:** *The* major problem. Fungal blights (Early & Late Blight), bacterial wilt, Tomato Yellow Leaf Curl Virus (TYLCV).
  - **Pests:** Whiteflies (spreads TYLCV), *Tuta absoluta* (Tomato Leaf Miner) is a devastating pest.
- **Problem:** Nigeria has a massive tomato deficit due to post-harvest losses and poor dry-season production.

## 6.3 PEPPER (*Capsicum spp.*)

- **Importance:** A key ingredient for flavour and heat.
- **Types in Nigeria:**
  - **Ata Rodo / Habanero (*C. chinense*):** Very hot, aromatic.
  - **Ata Jije / Bird's Eye (*C. frutescens*):** Small, very hot.
  - **Tatashe / Bell (*C. annuum*):** Mild, large, used for colour.
- **Cultivation:** Very similar to tomato (nursery, transplanting). They are generally *hardier* and more disease-resistant than tomatoes.

## 6.4 OKRA (*Abelmoschus esculentus*)

- **Importance:** A popular vegetable, used as a "thickener" for soups ('draw soup').
- **Cultivation:** Planted directly by seed.
- **Harvesting:** This is the *key point*. Pods must be harvested *every 2-3 days* while they are young and tender. If left on the plant, they quickly become **fibrous** and inedible. Regular harvesting also encourages the plant to produce more flowers.

## 6.5 LEAFY VEGETABLES (SPINACH, AMARANTH)

- **Types:**
  - **Amaranth (*Amaranthus spp.*):** "Efo Tete," "Green."
  - **Lagos Spinach / Fluted Pumpkin (*Telfairia occidentalis*):** "Efo Ugwu." (A vine).
  - **Celosia (*Celosia argentea*):** "Efo Shoko."
- **Cultivation:**
  - **Land Prep:** Planted on raised beds to ensure good drainage.
  - **Planting:** Seeds are tiny and are often "broadcast" (scattered) over the bed and lightly raked in.
  - **Fertilizer:** These are "leaf" crops, so they respond very well to **Nitrogen** (e.g., Urea or poultry manure) to promote lush leaf growth.
- **Harvesting:** Can be "once-over" (uprooting the whole plant) or "multiple-cut" (cutting the leaves, allowing the plant to re-sprout for 2-3 more harvests).

## EVALUATION (WEEK 6)

1. Why is tomato production often more successful in the *dry season*?
2. What is the advantage of raising tomato/pepper seedlings in a "nursery" first?
3. What is the *most important* management practice for harvesting Okra, and why?
4. Which *primary nutrient* (N, P, or K) do leafy vegetables like Amaranth require in large amounts, and why?
5. Name the 3 main species of pepper grown in Nigeria.

## ASSIGNMENT (WEEK 6)

1. Find out the full name of the "Ugwu" leaf (*Telfairia occidentalis*).
2. Unlike Amaranth, it is not grown from seed on a bed. How is it propagated, and how is it grown (i.e., does it climb or spread)? Write a short report.



## **WEEK 7: MIDTERM BREAK**

*This is a break from instruction. Students are encouraged to review the first half of the term's work.*

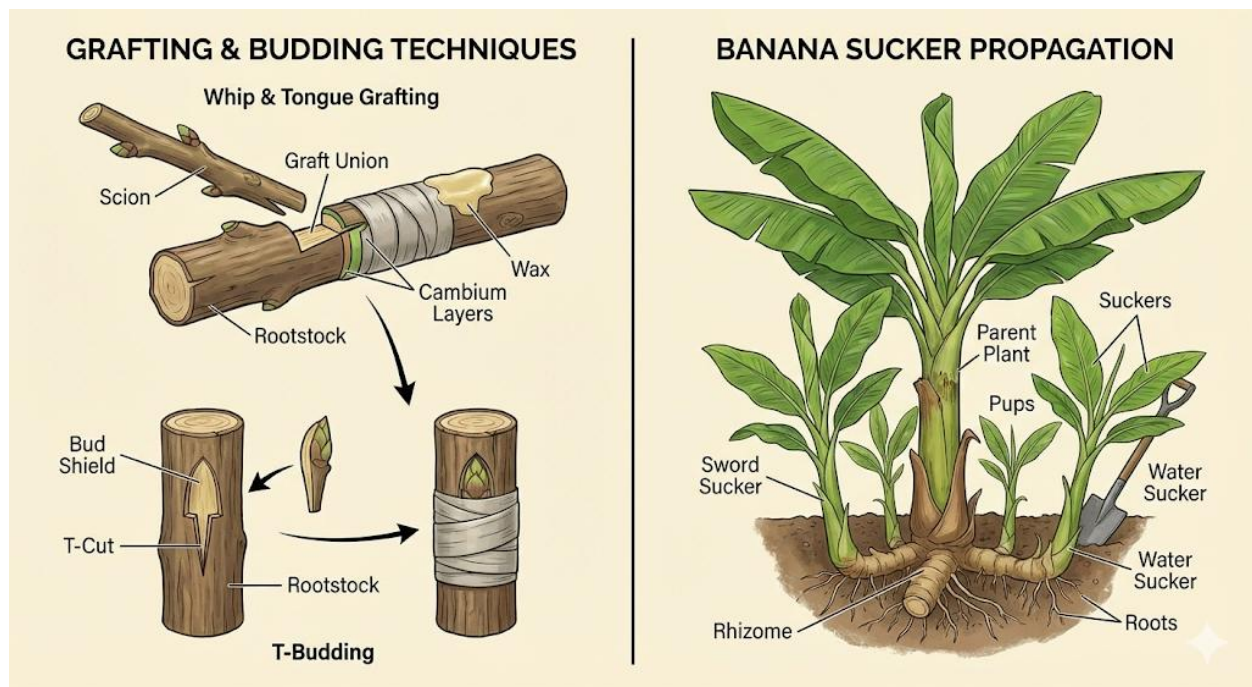




## WEEK 8: FRUITS

### 8.1 INTRODUCTION

Fruits are the mature, seed-bearing ovaries of flowering plants, typically eaten raw. They are sources of sugar (energy), vitamins (especially C and A), and fibre. They are high-value horticultural crops.



### 8.2 MANGO (*Mangifera indica*)

- **Importance:** A popular seasonal fruit.
- **Propagation:**
  - **Seed:** (Traditional) Easy, but the tree may not be "true-to-type" (like the parent) and takes long (7+ years) to fruit.
  - **Grafting/Budding:** (Modern/Commercial) This is the *best* method. A bud or branch from a superior tree (e.g., 'Kent', 'Keith') is joined onto a hardy "rootstock." This guarantees the fruit type and quality, and the tree fruits much faster (3-4 years).
- **Problems:**

- **Alternate Bearing:** A tendency to fruit heavily one year and very poorly the next.
- **Anthraxnose:** A fungal disease that causes black spots on leaves, flowers, and fruits, leading to rot.

### 8.3 CITRUS (*Citrus spp.*)

- **Importance:** Includes Sweet Orange (*C. sinensis*), Lemon (*C. limon*), Lime (*C. aurantifolia*), and Grapefruit (*C. paradisi*).
- **Propagation:**
  - **Budding (T-Budding):** This is the *standard commercial method*. A bud from the desired variety is inserted under the bark of a hardy rootstock (e.g., rough lemon). This ensures uniformity and disease resistance.
- **Cultivation:** A plantation crop, requires good soil, fertilizer, and pest control.

### 8.4 BANANA & PLANTAIN (*Musa spp.*)

- **Botany:** These are *giant herbs*, not trees. The "trunk" is a "pseudostem" made of rolled leaf sheaths.
- **Propagation:** They do not (normally) produce seeds. They are propagated vegetatively using "suckers" (also called "pups") that grow from the base of the mother plant.
- **Management:**
  - **De-suckering:** Removing excess suckers, leaving only 1 or 2 to replace the mother plant.
  - **Propping:** The fruit bunch is heavy. The plant must be "propped" with a forked stick to prevent it from falling over.
- **Pests & Diseases:** Banana Weevil, Black Sigatoka (a fungal leaf disease).

### 8.5 PINEAPPLE (*Ananas comosus*)

- **Propagation (Vegetative):**
  - **Suckers:** Grow from the base.
  - **Slips:** Grow from the stalk just below the fruit.
  - **Crown:** The leafy top of the fruit itself.

- **Cultivation:** A terrestrial (ground) plant. Planted on raised beds, often with black plastic mulch (to control weeds and warm the soil).
- **Harvest:** Takes 1.5 - 2 years from planting.

### EVALUATION (WEEK 8)

1. What is the *best* propagation method for commercial mango and citrus trees, and why is it better than planting a seed?
2. What is "Alternate Bearing" in mango?
3. How are bananas and plantains propagated?
4. What is "De-suckering" in plantain management?
5. List the three (3) materials that can be used to plant a pineapple.

### ASSIGNMENT (WEEK 8)

1. Buy a pineapple.
2. Twist off the "crown" (the leafy top).
3. Pull off the bottom 3-4 layers of small leaves to expose the "stem" base.
4. Place it in a cup of water (with the base submerged) and put it on a windowsill.
5. Observe it for 2 weeks. Write down what happens (e.g., "Small white roots began to grow"). This is you, propagating a pineapple.

## WEEK 9: SPICES AND CONDIMENTS

### 9.1 INTRODUCTION

Spices and condiments are substances added to food in small quantities to add flavour, aroma, colour, or "heat" (pungency). They are not eaten as a main course but are essential for making food palatable.





## 9.2 GINGER (*Zingiber officinale*)

- **Botany:** The part used is the **rhizome** (a fleshy, underground stem), *not* a root.
- **Ecology:** A tropical plant that needs a warm, humid climate, high rainfall, and *some shade*. It does well in forest or semi-forest conditions.
- **In Nigeria: Kaduna State** (Southern Kaduna) is the undisputed hub of ginger production.
- **Cultivation:**
  - **Planting:** Small pieces of the rhizome ("setts"), each with a few "eyes" or buds, are planted in beds or ridges.
  - **Mulching:** The beds are heavily mulched (with grass) to conserve moisture, keep the soil cool, and suppress weeds.
- **Harvest:** 8-10 months. The rhizomes are carefully dug up, washed, and dried. Sold fresh or dry.

## 9.3 GARLIC (*Allium sativum*)

- **Botany:** A member of the onion family. The part used is the **bulb**, which is composed of "cloves."
- **Ecology:** This is *not* a humid-tropical crop. It requires a **cool, dry season (Harmattan)** for the bulb to form properly.
- **In Nigeria:** Grown almost exclusively in the far North (e.g., Sokoto, Kano) during the *dry season* using irrigation.
- **Cultivation:** Individual "cloves" are separated and planted.

## 9.4 PEPPER (BLACK PEPPER) (*Piper nigrum*)

- **Important Distinction:** This is **not** the same as *Capsicum* (Ata rodo/Tatashe). This is the small, round peppercorn (Black, White, or Green pepper).
- **Botany:** It is a **climbing vine**.
- **Cultivation:** It must be "staked" or grown up a support (a pole or a living tree).
- **Products:**
  - **Black Pepper:** The *unripe* (green) fruit, harvested and *dried in the sun*, which makes it turn black and shrivel.

- **White Pepper:** The *ripe* (red) fruit, *soaked* in water (retting) to remove the outer flesh, leaving the white seed, which is then dried.

### 9.5 CLOVES (*Syzygium aromaticum*)

- **Botany:** Cloves are the **un-opened, dried flower buds** of a tropical evergreen tree.
- **Harvest:** The buds are hand-picked just before they open (when they turn pink) and are dried in the sun until they turn dark brown.

### EVALUATION (WEEK 9)

1. What part of the ginger plant is harvested and used?
2. Why is ginger production concentrated in Southern Kaduna (ecologically)?
3. Why is garlic production concentrated in the far North (ecologically)?
4. Explain the difference between *Capsicum* pepper (Tatashe) and *Piper nigrum* (Black Pepper).
5. What part of the plant are "Cloves"?

### ASSIGNMENT (WEEK 9)

1. Go to a kitchen or market.
2. Find *Black Pepper* and *White Pepper*.
3. Write a paragraph explaining that they both come from the *same plant*, and describe what process makes one black and the other white.



## WEEK 10: MEDICINAL PLANTS

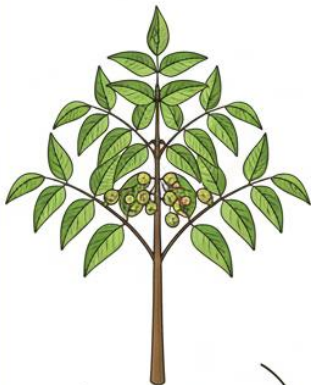
### 10.1 INTRODUCTION (ETHNOBOTANY)

Medicinal plants are plants that are used for their therapeutic properties to treat, cure, or prevent illness in humans and animals. This is the foundation of "Traditional Medicine." Many modern drugs are derived from compounds first discovered in plants.

## MEDICINAL PLANTS: NATURE'S PHARMANCY

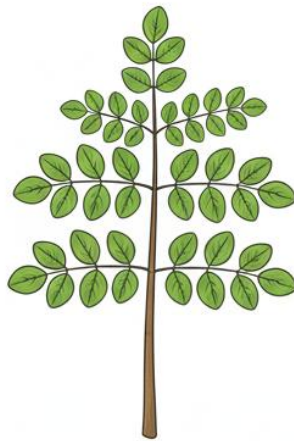
### NEEM

(Azadirachta indica)



### MORINGA

(Moringa oleifera)

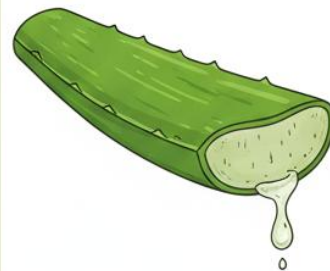


#### SUPERFOOD:

-  Vitamins A, C, E
-  Calcium & Iron
-  Protein

### ALEM VERA

(Morbalead miller)



#### SOOTHING GEL

Skincare & Healing

## 10.2 NEEM (*Azadirachta indica*)

- **Common Name:** "Dogonyaro" (Hausa).
- **Origin:** Native to India, but now widespread ("naturalized") all over Nigeria. It's a hardy, drought-tolerant tree.
- **Medicinal Uses:**
  - **Antimalarial:** The leaves and bark are boiled into a bitter tea (Agbo) to treat fever and malaria.
  - **Skin:** The oil and leaf-paste are used to treat skin diseases (e.g., eczema, rashes).
- **Horticultural Use (Very Important):**
  - **Natural Insecticide:** The seeds contain a powerful compound called **Azadirachtin**. This is extracted to make "**Neem Oil**," a safe, biodegradable insecticide used in organic farming to control pests.

## 10.3 MORINGA (*Moringa oleifera*)

- **Common Names:** "Miracle Tree," "Zogale" (Hausa).
- **Importance:** A fast-growing tree valued for its *incredible* nutritional and medicinal properties.
- **Medicinal/Nutritional Uses:**
  - **Nutrition:** The **leaves** are a "superfood." They are *packed* with protein, Vitamin A, Vitamin C, calcium, and potassium. Used to combat malnutrition.
  - **Water Purification:** The **seeds**, when crushed, act as a "flocculant." They cause dirt and microbes in water to clump together and sink, clarifying the water.
  - **Blood Pressure:** The leaves are used to make a tea to help regulate blood pressure.

## 10.4 ALOE VERA (*Aloe barbadensis*)

- **Botany:** A succulent plant that stores water in its thick, fleshy leaves.
- **Uses:**
  - **Skin (Topical Use):** The clear **gel** inside the leaf is a famous remedy for:
    - **Burns:** Cools and soothes fresh burns (sunburn, kitchen burns).

- **Skin Irritation:** Treats rashes, insect bites, and eczema.
- **Internal Use:** The **juice** (a different part, the yellow "latex") is a powerful laxative, used to treat constipation. *Must be used with care.*

### EVALUATION (WEEK 10)

1. What is the *active compound* in Neem seeds, and what is its main *horticultural* use?
2. Why is *Moringa* often called the "Miracle Tree"? List two of its uses.
3. Explain how Moringa seeds can be used to purify water.
4. What is the primary medicinal use of the *gel* from an Aloe Vera leaf?

### ASSIGNMENT (WEEK 10)

1. Safely (with permission) get one leaf of Aloe Vera.
2. Cut it open. Observe and describe the clear "gel."
3. Research the difference between the "clear gel" (used for skin) and the "yellow latex" (used as a laxative) in the Aloe leaf. Write a paragraph explaining why you should not just eat the whole leaf.

## WEEK 11: REVISION

- **Review of Key Concepts:**

- **Week 1 (Cereals):** Maize (NPK, Fall Armyworm), Rice (Upland vs. Lowland, Parboiling), Sorghum/Millet (Drought).
- **Week 2 (Legumes):** N-Fixation. Cowpea (Pest complex, PICS bags), Soybean (Shattering, Inoculation), Groundnut (Geocarp, Aflatoxin).
- **Week 3 (Tubers):** Cassava (Hardy, Cyanide, Processing, Perishable), Yam (Staking, Mounds, Barn storage), Cocoyam (Shade/Water), Potato (Cool climate).
- **Week 4 (Oils):** Oil Palm (Tenera, Palm Oil vs. PKO), Coconut (Copra), Sunflower (Annual).
- **Week 6 (Veg):** Tomato (Nursery, Dry season), Okra (Harvest frequency), Leafy (N-fertilizer).
- **Week 8 (Fruits):** Propagation (Grafting/Budding vs. Suckers), Mango (Alternate bearing), Banana (De-suckering).
- **Week 9 (Spices):** Ginger (Rhizome, Kaduna), Garlic (Bulb, North), Pepper (*Piper nigrum*), Cloves (Flower bud).
- **Week 10 (Medicinal):** Neem (Insecticide), Moringa (Nutrition, Water), Aloe (Skin gel).

## EVALUATION (WEEK 11)

1. Compare and contrast the *planting material* used for Yam, Cassava, and Rice (transplanted).
2. A farmer has a field in Jos and a field in Calabar. Which locations are suitable for: (a) Irish Potato, (b) Oil Palm, (c) Garlic, (d) Yam?
3. What is the key difference between the propagation of Citrus and the propagation of Plantain?
4. Aflatoxin is a major problem in which Week 2 crop? *Tuta absoluta* is a pest of which Week 6 crop? Fall Armyworm is a pest of which Week 1 crop?